

JACC STATE-OF-THE-ART REVIEW

Biomarkers for Cardiovascular Drug Development



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ABSTRACT

Circulating and urinary laboratory biomarkers are central to cardiovascular, kidney, and metabolic drug development, and they support diagnosis, risk stratification, dose finding, safety monitoring, and, in selected settings, endpoint definition. Yet only a few materially shape trial design or regulatory decisions, and still fewer meet stringent criteria for surrogate endpoints. This review proposes a context-of-use framework in which circulating biomarkers are assessed along 2 converging “evidence lanes”: analytical readiness and clinical or statistical validation. The review outlines fit-for-purpose requirements for enrichment tools, pharmacodynamic and dose-ranging markers, safety biomarkers, and candidate surrogates, by emphasizing preanalytical, analytical, and postanalytical sources of variability. Particular attention is given to surrogate validation, highlighting frequent discordance between biological plausibility or prognostic strength and treatment-induced changes that reliably predict clinical benefit (eg, albuminuria, natriuretic peptides, and glycemic, inflammatory, and lipid markers). The review describes how biomarkers can be embedded prospectively in phase 1 through phase 4 trials to guide dose selection, enrich populations, trigger adaptive design decisions, and support postmarketing surveillance while generally retaining hard outcomes as the basis for efficacy claims. A dedicated section summarizes perspectives from major regulatory agencies, and another applies the framework to concrete cardiovascular examples, illustrating both successful and unsuccessful uses. Finally, the review discusses practical barriers—analytical variability, operational burden, and limited commercial incentives—and proposes shared solutions, including standardized assays, harmonized data standards, open access platforms, and cross-trial biobanks. (JACC. 2026;87:1950–1972) © 2026 by the American College of Cardiology Foundation.

Biomarkers are measurable indicators of a physiological state or pathologic process. They may take many forms, ranging from blood-derived molecules and genetic variants to urinary substances, imaging results, and other molecular signatures, each offering unique insights into disease pathogenesis, progression, and response to

treatment. As such, biomarkers have become integral to modern clinical drug development, particularly as the field moves toward personalized and precision medicine.¹

Circulating and urinary biomarkers have assumed particular importance in cardiovascular (CV) research and clinical trials.² These molecules

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HIGHLIGHTS

- Circulating biomarkers support trial design by informing diagnosis, risk stratification, dose selection, and safety.
- Most biomarkers fail to change decisions because of weak treatment linkage or lack of outcome correlation.
- Pharmacodynamic and dose-ranging biomarkers can enhance early-phase development and adaptive trial design.
- Surrogate endpoints require within-trial evidence showing that biomarker changes predict patient benefit.
- Standardized assays, reference change values, and prospective validation are essential for clinical and regulatory impact.

can be sampled at enrollment and throughout study procedures, thus providing real-time insight into disease processes and drug effects. By revealing subtle pathophysiological changes before clinical manifestations and identifying patients who are more likely to respond to therapy, laboratory biomarkers are poised to streamline the drug development process; as an example, in trials, changes in biomarkers are increasingly held to represent evidence of treatment efficacy. Meanwhile, advances in multiomics platforms continue to reveal new candidate biomarkers, further enhancing our ability to stratify CV risk and monitor disease progression. As clinical research progresses toward fully individualized medicine, biomarker-guided methodologies will play a pivotal role in improving patient outcomes and expediting treatment development.^{1,3}

A discussion about biomarker-guided drug development in the CV field was initiated during the 2024 Global Cardiovascular Clinical Trials Forum in Washington, DC. Building on that dialogue, this report explores how biomarkers have transformed CV research, offers strategies for their effective implementation, and outlines future directions for innovation. In this review, unless otherwise specified, we use *biomarkers* to refer to circulating and urinary laboratory markers; other modalities (imaging, device diagnostics, electrocardiographic/digital signals) are acknowledged but lie outside of our scope.

POSSIBLE APPLICATIONS OF BIOMARKERS

Biomarkers can be classified into 5 main categories according to their practical use: diagnostic, prognostic, pharmacodynamics (response) and dose-ranging, safety, and surrogate endpoints (Table 1). Surrogate endpoints are discussed separately.

DIAGNOSTIC BIOMARKERS. Diagnostic biomarkers are valuable for confirming the presence and severity of disease and for elucidating underlying mechanisms. They support clinical decision making, guide the development of targeted therapies, and help identify patients most likely to benefit. In some settings, biomarkers can also help rule out disease; an example is cardiac troponin measurement in suspected acute myocardial infarction.⁴ Other diagnostic biomarkers include N-terminal pro-B-type natriuretic peptide (NT-proBNP) for suspected heart failure (HF),⁵ glycated hemoglobin for diabetes,⁶ and the urine albumin-to-creatinine ratio (UACR) for kidney impairment.⁷

PROGNOSTIC BIOMARKERS. Prognostic biomarkers provide insight into individual patient risk. Incorporating these indicators into trial design improves risk stratification, thereby enabling smaller, more efficient studies with clearer efficacy signals. For example, natriuretic peptides (NPs) have been most widely used to enrich trials (eg, DAPA-HF [Study to Evaluate the Effect of Dapagliflozin on the Incidence of Worsening Heart Failure or Cardiovascular Death in Patients With Chronic Heart Failure; NCT03036124], which required NT-proBNP ≥ 400 ng/L⁸) and to avoid enrolling patients at such high risk that a meaningful treatment response is unlikely (eg, STRONG-HF [Safety, Tolerability and Efficacy of Rapid Optimization, Helped by NT-proBNP testinG, of Heart Failure Therapies; NCT03412201], which required a $>10\%$ decrease in NT-proBNP between screening and randomization⁹).

PHARMACODYNAMIC (RESPONSE) AND DOSE-RANGING BIOMARKERS

Pharmacodynamic biomarkers provide direct evidence of a biological response to therapeutic intervention and, when prospectively integrated into exposure-response analyses, can also function as dose-ranging tools. In early-stage development, they help confirm that a therapy is on target, inform dose selection, and indicate whether

ABBREVIATIONS AND ACRONYMS

CV	= cardiovascular
EMA	= European Medicines Agency
FDA	= U.S. Food and Drug Administration
HF	= heart failure
HFpEF	= heart failure with preserved ejection fraction
hs-CRP	= high-sensitivity C-reactive protein
hs-cTn	= high-sensitivity cardiac troponin
hs-cTnI	= high-sensitivity cardiac troponin I
hs-cTnT	= high-sensitivity cardiac troponin T
LDL-C	= low-density lipoprotein cholesterol
NP	= natriuretic peptide
NT-proBNP	= N-terminal pro-B-type natriuretic peptide
SGLT2	= sodium-glucose cotransporter 2
UACR	= urine albumin-to-creatinine ratio

TABLE 1 Categories of Circulating Biomarkers in CV Drug and Device Development

Category	Definition	Typical Context of Use	Phase-Specific Roles	Acceptability Notes (Regulatory Stance)	Examples (CV Drug/Device)
Diagnostic	Confirm presence of disease or characterize subtype/severity; support eligibility and differential diagnosis.	Screening; entry criteria; baseline characterization; adjudication support in composites (where specified)	Phase 1-2: baseline characterization Phase 3: confirm inclusion/exclusion and stratification Phase 4: surveillance and phenotyping	Supportive (not approvable for efficacy); central/standardized assays preferred for trial consistency	cTn for AMI; NT-proBNP for suspected HF; HbA _{1c} (diabetes context); UACR for kidney impairment
Prognostic	Estimate risk and likely clinical trajectory independent of treatment assignment.	Risk enrichment; stratified randomization; subgroup analyses	Phase 2: enrich to raise event rates Phase 3: stratification and prespecified heterogeneity analyses Phase 4: risk tools in registries	Supportive for design/interpretation; not an efficacy endpoint	NT-proBNP (HF prognosis), hs-cTn (incident CHD/HF), hs-CRP (inflammatory risk)
Pharmacodynamic (response) and dose-ranging	Demonstrate treatment-induced biological change; characterize exposure response to inform dose or device parameters.	Proof of mechanism; dose/parameter selection; adaptive rules; early go/no-go	Phase 1-2: primary decision tools for dose-/parameter-ranging with pre-specified targets and RCV Phase 3: supportive (eg, key secondary), link to outcomes Phase 4: therapy optimization and monitoring	Generally supportive unless validated as surrogates; centralization and predefined RCV recommended	LDL-C with lipid-lowering therapies; Lyso-Gb3 in Anderson-Fabry; NT-proBNP (mechanism-specific PD in HF); UACR for cardiorenal mechanisms; LDH/plasma-free hemoglobin for blood-contacting devices
Safety	Identify or predict treatment-related harm; monitor organ injury/toxicity or device-specific complications.	Risk management plans; stopping boundaries; signal detection in phases 1-4	Phase 1-2: early safety thresholds and adjudication Phase 3: ongoing monitoring with predefined actions Phase 4: pharmacovigilance and device surveillance	Widely accepted for safety monitoring; not efficacy claims; consideration of drug-induced biomarker changes unrelated to toxicity (eg, <i>NPPB</i> derepression)	ALT/AST (liver), creatinine/eGFR (kidney), neutrophil count; hs-cTn for cardiotoxicity; LDH/plasma-free hemoglobin (hemolysis) for LVADs/valves; D-dimer for thrombosis risk.
Surrogate endpoints	Substitute for patient-important outcomes when validated or deemed reasonably likely to predict benefit for a given mechanism and setting.	Efficacy endpoints in select contexts; otherwise, key secondary with outcomes backstop	Phase 2: exploratory or mechanism-specific (often PD); may inform accelerated pathways if “reasonably likely” with confirmatory outcomes Phase 3: traditional approval requires validated surrogates or clinical outcomes Phase 4: durability and label refinement	FDA/EMA tiers: validated (traditional approval), reasonably likely (accelerated approval with confirmatory trials), supportive only (enrichment/PD)	Validated: LDL-C (lipid-lowering) Supportive/under evaluation: NT-proBNP in HF (enrichment/PD), eGFR slope (context-dependent) Glycemic: HbA _{1c} (glycemic control; not a CV surrogate for many mechanisms)

ALT = alanine aminotransferase; AMI = acute myocardial infarction; AST = aspartate aminotransferase; CHD = coronary heart disease; cTn = cardiac troponin; CV = cardiovascular; D-dimer = fibrin degradation product; eGFR = estimated glomerular filtration rate; EMA, European Medicines Agency; FDA = U.S. Food and Drug Administration; HbA_{1c} = hemoglobin A_{1c}; HF = heart failure; hs-CRP = high-sensitivity C-reactive protein; hs-cTn = high-sensitivity cardiac troponin; LDH = lactate dehydrogenase; LDL-C = low-density lipoprotein cholesterol; LVAD = left ventricular assist device; Lyso-Gb3 = lysoglobotriaosylceramide; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PD = pharmacodynamic; RCV = reference change value; UACR = urine albumin-to-creatinine ratio.

larger, later-stage trials are warranted. Examples include low-density lipoprotein cholesterol (LDL-C) with lipid-lowering therapies,^{10,11} globotriaosyl-sphingosine in Anderson-Fabry disease,¹² short-term NT-proBNP trajectories in HF for agents with neuro-hormonal or hemodynamic effects,^{13–16} and UACR for cardiorenal mechanisms.^{17–19} Biomarkers intended for dose selection should meet several criteria²⁰: 1) they are mechanistically on pathway for the drug’s primary effect; 2) they exhibit a discernible, preferably monotonic, relationship to exposure across the anticipated clinical range; 3) they have low biological variation, with a predefined reference change value to distinguish signal from noise; 4) they are

measured using analytically robust, preferably centralized, assays; and 5) they respond on a time scale compatible with iterative decision making. Early cohorts should include pharmacokinetic or pharmacodynamic sampling with a prespecified quantitative pharmacodynamic target (eg, achieving a large LDL-C reduction by 8–12 weeks for proprotein convertase subtilisin/kexin type 9 [PCSK9]-pathway therapies or ~20%–30% NT-proBNP reduction within 4 to 12 weeks for suitable HF mechanisms).^{10,11,13,15,16} Serial measurement enables adaptive trial designs,²¹ thus allowing dose titration or design modifications as evidence accumulates. Class-specific biology matters. Some effective

therapies show modest or delayed effects on certain pharmacodynamic markers (eg, sodium-glucose cotransporter 2 [SGLT2] inhibitors and natriuretic peptides [NPs]²²), whereas others can increase pharmacodynamic markers through off-target effects—for example, antiandrogens increasing NPs through derepression of *NPPB*, in a way that is unrelated to toxicity.²³⁻²⁵ Accordingly, pharmacodynamically guided dose decisions should be based on multiparametric evaluation (including imaging and functional measures) and interpreted in the specific clinical and mechanistic context.

SAFETY BIOMARKERS. Traditional safety biomarkers include laboratory measures of kidney and liver function, as well as blood counts, particularly neutrophil count.²⁶ Cardiac biomarkers are increasingly important for monitoring therapy-related cardiotoxicity; for example, high-sensitivity cardiac troponin I (hs-cTnI) and high-sensitivity cardiac troponin T (hs-cTnT) are used as safety markers in patients with breast cancer who are receiving trastuzumab.²⁷ Importantly, any safety program must distinguish drug-induced changes in biomarker levels that reflect pharmacologic activity or adaptive physiology from those that indicate true organ toxicity. Regulatory guidance on biomarker qualification explicitly states that treatment-emergent biomarker changes may either predict safety problems or reveal pharmacologic activity expected to predict clinical benefit, and that the same biomarker can serve as a safety signal in one setting and a desired pharmacodynamic effect in another (eg, blood pressure, glomerular filtration rate, serum lipids).²⁸ Clear fit-for-purpose criteria are therefore needed before a marker is relied on as a validated safety biomarker, including analytical robustness in the relevant decision range, predefined reference change values, characterization of important modifiers, and centralized or harmonized measurement.

Recent phase 2 HF trials illustrate how safety biomarkers can provide early signals of potential harm and alter development trajectories. In a study of sodium zirconium cyclosilicate to facilitate mineralocorticoid receptor antagonist use in HF with reduced ejection fraction, trial investigators and independent commentators observed higher NT-proBNP levels and a numerical excess of HF hospitalizations in the active-treatment arm (composite of CV death or worsening HF in 11 [11%] participants in the sodium zirconium cyclosilicate group and 3 [3%] in the placebo group; $P = 0.034$), raising concerns about fluid retention effects related to sodium load.²⁹ In heart failure with preserved ejection

fraction (HFpEF), the long-acting relaxin analogue volenrelaxin was associated with worsening congestion and increases in NT-proBNP, with signals toward more HF events; the sponsor stopped the trial early after an interim analysis indicated a low probability of benefit, thereby averting exposure of larger groups in phase 3 studies.³⁰ Together, these examples underscore the value of prespecified safety biomarker thresholds and adjudication plans: when biomarkers move in a direction inconsistent with benefit—and especially when aligned with adverse trends—programs should be equipped to pause, adapt, or stop to protect patients and conserve resources.

ROLES OF BIOMARKERS IN DIFFERENT PHASES OF DRUG/DEVICE DEVELOPMENT

Biomarkers play a pivotal role throughout the development continuum for both drugs and devices (Table 2).

PRECLINICAL AND PHASE 1 (FIRST-IN-HUMAN/FEASIBILITY). Biomarkers are essential for target validation and for assessing pharmacodynamic effects. Mechanistic biomarkers can signal early biological responses, thereby guiding dose selection and supporting the transition from animal models to human trials. For example, in the phase 1 CRISPR-Cas9 In Vivo Gene Editing for Transthyretin Amyloidosis trial, nexiguran ziclumeran reduced transthyretin levels by up to 90% at 12 months after a single administration,^{31,32} informing a phase 3 clinical trial (A Phase 3, Multinational, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of NTLA-2001 in Participants With Transthyretin Amyloidosis With Cardiomyopathy [MAGNITUDE]; NCT06128629). For devices, analogous principles apply, although the emphasis is on imaging and tissue-response biomarkers. First-in-human feasibility often relies on procedural success and injury biomarkers such as high-sensitivity cardiac troponin (hs-cTn) after ablation or structural interventions, hemolysis markers for blood-contacting devices, D-dimer for thrombosis risk, and early renal or hepatic panels for off-target effects.

PHASE 2 (OPTIMIZATION, SIGNAL DETECTION, AND ENRICHMENT). Phase 2 drug trials often use biomarkers to refine patient selection and monitor treatment response in real time, with an emphasis on mechanistic associations and dose refinement. This approach increases sensitivity to detect efficacy signals, helps identify early-response biomarkers, and can guide go/no-go decisions. For devices, phase 2

TABLE 2 Biomarker Applications Across Drug Development Phases

Phase/Setting	Diagnosis	Risk Prediction (Prognostic)	Pharmacodynamic/Dose-Ranging	Safety	Surrogate Endpoints
Preclinical and translational	Phenotyping in models; candidate human matrices/units; define eligibility assays for later phases	Risk biology mapping; natural history cohorts to define baseline distributions	PK/PD relationships; exposure-response windows; assay feasibility and stability	Toxicology biomarkers in vivo/ex vivo; organ-injury panels; platform stability checks	Candidate signals only (not surrogates); specify hypotheses for later validation
Phase 1 (first-in-human)	Baseline phenotyping; confirm inclusion/exclusion with standardized assays	Exploratory baseline tagging of risk (eg, NP, hs-cTn) to inform later enrichment	Dose escalation with PD windows; early target engagement using responsive markers	DLT and cardiac safety: ECG/QTc, hs-cTn; liver/kidney panels with rapid turnaround	Exploratory change signals only; not used for efficacy claims
Phase 2 (proof-of-concept/optimization)	Standardized eligibility thresholds (eg, NP ranges in HF, UACR in CKD)	Risk enrichment (eg, NP $\geq$ cutoff; UACR $\geq$ range) stratified randomization by risk or biomarker	Dose-/parameter-ranging guided by PD targets (eg, LDL-C fraction of Emax; NT-proBNP at 4–12 wk when mechanism-appropriate)	Safety panels with trigger-based holds/adjustments; monitor for discordant NP rises with adverse trends (eg, HF signals)	Intermediate markers (eg, NT-proBNP, LDL-C) inform design; surrogacy claims generally premature
Phase 3 (pivotal)	Eligibility thresholds harmonized across sites; central adjudication where assays feed composites	Prespecified strata and subgroup analyses; maintain generalizability while preserving event rates	On-treatment PD monitoring; intercurrent event handling tied to biomarker context of use	Event-driven safety monitoring; adjudication and DSMB; centralized/standardized laboratories	Validated surrogates only (eg, LDL-C in lipid-lowering); co-primary/hierarchical schemes with outcomes backstops when applicable
Phase 4 (postmarketing)	Registry phenotyping; real-world assay performance and reference intervals	Risk tools in registries; label expansion signals across subgroups and practice settings	Treatment-response monitoring; remote/device diagnostics integrated with laboratory tests	Pharmacovigilance; signal detection for harm and drift; firmware/software update surveillance (devices)	Effectiveness in routine care (if any); real-world evidence to support or refute surrogate utility

Table showing how biomarker roles (diagnosis, risk prediction, pharmacodynamic/dose-ranging, safety, surrogate) evolve from preclinical work through phases 1 to 4: from defining matrices and PK/PD windows, to enrichment and dose optimization, to restricted surrogate use in pivotal trials, and finally to real-world risk tools and safety surveillance.
CKD = chronic kidney disease; DSMB = data and safety monitoring board; DLT = dose-limiting toxicity; Emax = maximal effect; NP = natriuretic peptide; PK = pharmacokinetic; QTc = corrected QT interval; other abbreviations as in Table 1.

trials typically emphasize optimization of device parameters and user-device interaction; circulating biomarkers more often serve as enrichment criteria (eg, NT-proBNP or functional/imaging markers to select candidates for cardiac resynchronization therapy or valvular interventions).

PHASE 3 (CONFIRMATORY EFFICACY AND SAFETY). In phase 3 drug trials, biomarkers complement traditional clinical endpoints by providing mechanistic insight and supporting the overall assessment of efficacy and safety. In some cases, they can act as surrogate endpoints for predicting meaningful clinical outcomes. For devices, phase 3 (often pivotal) trials frequently combine clinical outcomes with biomarker-anchored secondary endpoints to confirm a durable mechanism (eg, NT-proBNP after structural or hemodynamic devices). Safety monitoring incorporates device-relevant biomarkers (eg, hemolysis for ventricular assist devices or valves; myocardial or renal injury after structural heart procedures).

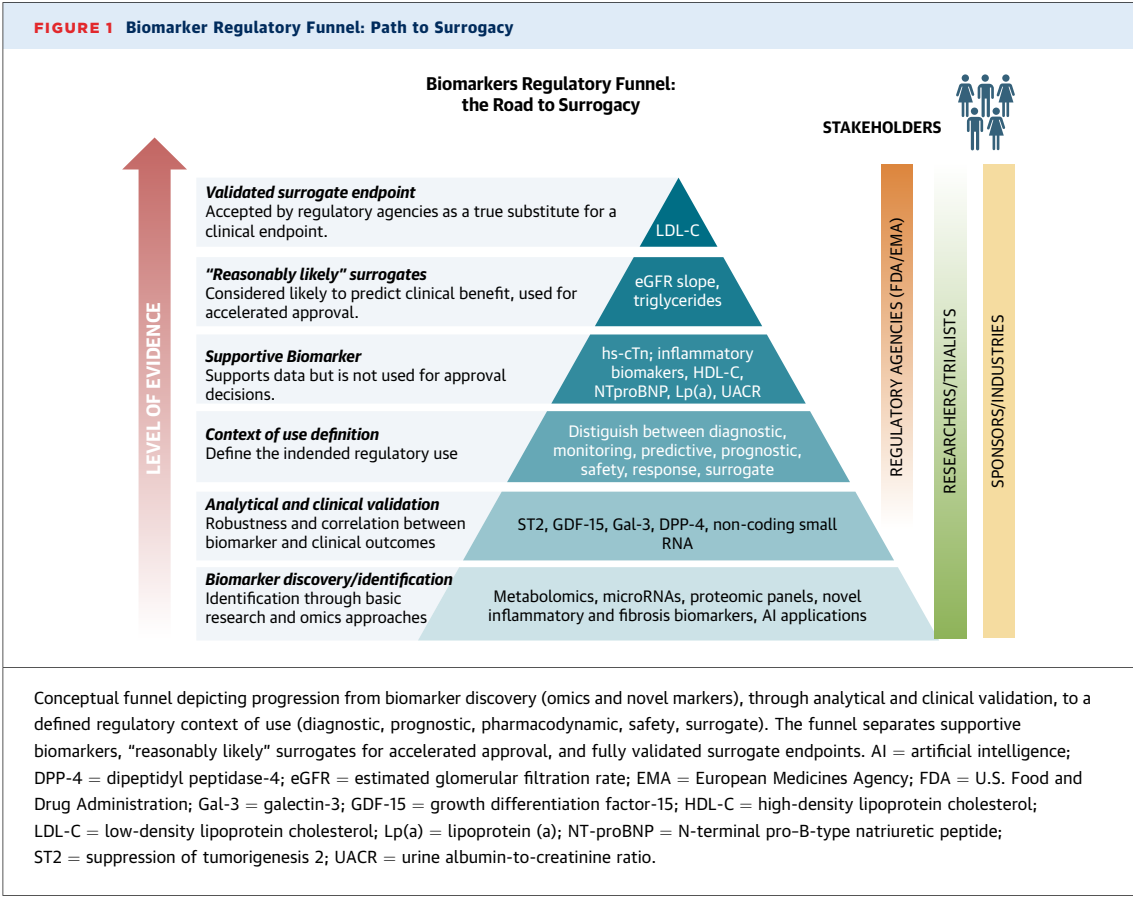
PHASE 4 (POSTMARKETING, LIFE CYCLE MANAGEMENT). Biomarkers can inform long-term monitoring and safety surveillance and help demonstrate real-world

effectiveness. For devices, longitudinal laboratory biomarkers complement device diagnostics to track durability of effect.

BIOMARKERS AS SURROGATE ENDPOINTS: PEARLS AND PITFALLS

A surrogate endpoint is a clinical trial endpoint used as a substitute for a direct measure of how a patient feels, functions, or survives. To be useful, a surrogate must correlate with the true clinical outcome and fully capture the net effect of treatment on that outcome.³³ Circulating biomarkers may play a role as surrogate endpoints, but major expectations must be met.

The value of circulating biomarkers as surrogate endpoints is clear; such surrogates can shorten development timelines when hard clinical outcomes accrue slowly, but they add value only when they are tightly linked to disease biology and when treatment-induced changes in the surrogate reliably translate into changes in clinical events (Figure 1). A classic success story in CV drug development is LDL-C: it lies on the causal pathway of atherosclerosis, shows a roughly monotonic exposure-response relationship



(greater LDL-C reductions yield greater risk reduction), and decades of randomized trials have confirmed that drug-mediated LDL-C lowering consistently reduces CV events. By contrast, interventions aimed at raising high-density lipoprotein cholesterol have not improved patient outcomes despite favorable changes in the biomarker; accordingly, high-density lipoprotein cholesterol is not accepted as a surrogate endpoint for CV benefit.³⁴ Similarly, novel HF biomarkers such as galectin-3 or soluble ST2, albeit clearly predictive of worse outcomes, have not yet demonstrated treatment-responsive changes that consistently forecast clinical benefit across different therapeutic mechanisms. In this regard, in a recent analysis of 30 candidate biomarkers in HF, only changes in NT-proBNP independently predicted a subsequent risk of death or hospitalization after therapy adjustment.³⁵ NT-proBNP therefore occupies an intermediate position: it has a strong biological rationale, it is useful for risk enrichment, and evidence linking short-term NT-proBNP shifts to differences in outcomes is growing.¹³ Nevertheless, the role of NT-proBNP as a

surrogate outcome is not fully established across all contexts, so hard clinical outcomes remain the definitive standard for demonstrating benefit.¹³

Regulatory agencies such as the U.S. Food and Drug Administration (FDA) explicitly distinguish between validated surrogates (proven across diverse trials to predict clinical benefit, used for traditional approvals) and "reasonably likely" surrogates, which may support accelerated approval but require confirmatory outcome trials.³⁶ Regulatory aspects of surrogate use in CV drug development are discussed in more detail in a subsequent section.

Surrogate biomarkers can be applied at multiple stages of CV drug development. In phase 2 trials, they are typically used with the expectation that defined changes in the biomarker will anticipate the findings of a subsequent definitive phase 3 study.¹³ However, strong evidence is required even for this decision-making role. For example, NPs are frequently used in this setting, but a trial-level analysis of 16 phase 3 chronic HF trials found that treatment-related changes in NPs were not correlated with longer-term treatment effects on all-cause mortality

TABLE 3 Examples of Surrogate Endpoints in Phase 3 CV Trials

Disease Context	Biomarker (Surrogate)	Biological Rationale	Outcome Evidence	Surrogacy Outcome	Key Trials and Years
ASCVD prevention (LDL-lowering)	LDL-C reduction	Causal in atherogenesis; strong monotonic exposure-response	Reduced CV events across multiple mechanisms	Validated	CTT, 2010; IMPROVE-IT, 2015; FOURIER, 2017
Dyslipidemia (HDL-raising)	HDL-C increase	Not reliably on-pathway; not functional	No CV event reduction; off-target toxicity observed	Failed	AIM-HIGH, 2011; HPS2-THRIVE, 2014; ILLUMINATE, 2007; ACCELERATE, 2017
Chronic heart failure (HFrEF)	NT-proBNP % change (short-term)	Reflects hemodynamic load; responsive to treatment	Promising in trials but not consistently predictive of outcomes	Uncertain / Supportive	PROVE-HF, 2019; EVALUATE-HF, 2019
Cardiorenal programs	UACR (albuminuria) reduction	On-pathway for renal protection	Associated with renal benefit; CV linkage mechanism-dependent	Validated (renal)/ Uncertain (CV)	CREDENCE, 2019; DAPA-CKD, 2020
Inflammation and residual CV risk	hs-CRP reduction	Marker, possibly not causal mediator	Mixed results; agent- and pathway-specific benefits	Uncertain	JUPITER, 2008; CANTOS, 2017
Lipoprotein(a)-targeting	Lp(a) reduction	Genetic support for causality	Awaiting outcome validation	Pending	Tsimikas, 2020; OCEAN(a)-DOSE, 2022
Triglyceride-lowering	TG reduction	Plausibly on-pathway, but inconsistent across agents	REDUCE-IT showed benefit; STRENGTH and ACCORD-Lipid did not	Uncertain	REDUCE-IT, 2018; STRENGTH, 2020; ACCORD-Lipid, 2010
Fabry cardiomyopathy	Lyso-Gb3 reduction	Specific for Fabry disease; responsive to treatment	Used for monitoring; not yet validated as surrogate	Supportive/ monitoring only	Various Fabry trials (ERT, migalastat)

Illustrative Phase 3 programs in which circulating biomarkers have been used or proposed as surrogate endpoints, with their biological rationale, outcome evidence, and current surrogacy status.

ACCELERATE = Assessment of Clinical Effects of Cholesteryl Ester Transfer Protein Inhibition With Evacetrapib in Patients at a High Risk for Vascular Outcomes; ACCORD-Lipid = Action to Control Cardiovascular Risk in Diabetes; AIM-HIGH = Atherothrombosis Intervention in Metabolic Syndrome With Low HDL/High Triglycerides: Impact on Global Health Outcomes; ASCVD = atherosclerotic cardiovascular disease; CANTOS = Cardiovascular Risk Reduction Study [Reduction in Recurrent Major CV Disease Events]; CREDENCE = Canagliflozin and Renal Events in Diabetes With Established Nephropathy Clinical Evaluation; CTT = Cholesterol Treatment Trialists' Collaboration; CV = cardiovascular; DAPA-CKD = Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease; ERT = enzyme replacement therapy; EVALUATE-HF = Study of Effects of Sacubitril/Valsartan vs Enalapril on Aortic Stiffness in Patients With Mild to Moderate HF With Reduced Ejection Fraction; FOURIER = Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk; HDL-C = high-density lipoprotein cholesterol; HFrEF = heart failure with reduced ejection fraction; HPS2-THRIVE = Heart Protection Study 2-Treatment of HDL to Reduce the Incidence of Vascular Events; ILLUMINATE = Investigation of Lipid Level Management to Understand its Impact in Atherosclerotic Events; IMPROVE-IT = Improved Reduction of Outcomes: Vytorin Efficacy International Trial; JUPITER = Crestor 20 mg Versus Placebo in Prevention of Cardiovascular Events; Lp(a) = lipoprotein(a); OCEAN(a)-DOSE = Olpasiran Trials of Cardiovascular Events and Lipoprotein[a] Reduction-Dose Finding Study; PROVE-HF = Effects of Sacubitril/Valsartan Therapy on Biomarkers, Myocardial Remodeling, and Outcomes; REDUCE-IT = Reduction of Cardiovascular Events with Icosapent Ethyl-Intervention Trial; STRENGTH = Long-Term Outcomes Study to Assess Statin Residual Risk with Epanova in High Cardiovascular Risk Patients with Hypertriglyceridemia; TG = triglyceride; UACR = urine albumin-to-creatinine ratio; other abbreviations as in Table 1.

($r = 0.12$; $P = 0.63$), whereas they were correlated with effects on HF hospitalization ($r = 0.63$; $P = 0.008$).¹³ Correlation with HF hospitalization strengthened when analyses were restricted to studies using NT-proBNP assays, to results from more recent trials (>2010), and to investigations of inhibitors of the renin-angiotensin-aldosterone system.¹³ These findings underscore both the potential and the limitations of NPs as surrogates but are hardly surprising: for change in a biomarker to be tightly linked to effect of a therapy, the biomarker assays used should be robust, the therapy should have a meaningful impact on HF outcomes (reflected in biomarker changes), and the biology of the therapy should intersect with the biology of the biomarker used.

In phase 3 trials, the use of validated surrogate biomarkers is highly selective. Most biomarkers serve as supportive evidence, and only a subset of these biomarkers are considered likely to predict clinical benefit.³⁷ Candidate surrogates are commonly used as secondary endpoints and to guide midtrial decisions (Supplemental Table 1). As one FDA guideline notes, enrichment or surrogate endpoints can

improve efficiency, but evaluation of hard clinical outcomes is essential to confirm that translating a biomarker effect into patient benefit has been successful.³⁸ Examples of surrogate endpoints in phase 3 trials are summarized in Table 3.

Another phase 3 strategy is to specify a surrogate-anchored endpoint as a key secondary outcome (or occasionally as a component of a coprimary endpoint) within a hierarchical testing scheme that prioritizes clinical outcomes. In this framework, even if the surrogate shows a favorable treatment effect, overall trial success generally requires a benefit on the primary clinical endpoint. The surrogate then provides supportive evidence rather than carrying the full weight of regulatory approval. Sponsors often prespecify a surrogate “threshold effect” (ie, a hypothesized magnitude of biomarker change considered clinically meaningful) and then test whether observed biomarker changes in the treatment arm track with outcome benefits at both the trial and individual-patient levels. Such surrogate validation analyses examine whether patients or groups with larger biomarker improvements also experience

better clinical outcomes, thus strengthening the case that the surrogate reflects the treatment's effect on the underlying disease process.

In phase 4 (postmarketing), registries and real-world data can be used to examine whether treatment-associated biomarker changes predict and correlate with long-term clinical outcomes. For instance, a large analysis combining real-world HF data from >100,000 patients with data from 20 clinical trials found a consistent relationship between NT-proBNP levels and subsequent mortality or hospitalization risk in both settings, and the analysis showed that short-term changes in NT-proBNP could predict the longer-term outcome results of phase 3 trials.³⁹ Another example comes from nephrology: meta-analyses have demonstrated that a 30% reduction in albuminuria over 6 months predicts roughly a 30% reduction in long-term risk of kidney failure.⁴⁰ On the basis of such evidence, postmarket studies can use early changes in albuminuria to project clinical benefit.

Despite the potential strengths of biomarkers as a surrogate outcome measure, therapy developers and trialists must recognize the limits of surrogate-based expansion. If a new patient group, disease stage, or therapeutic class lacks evidence that changes in the surrogate translate into improved outcomes, relying on the surrogate to broaden use is risky. Furthermore, understanding that surrogates are most likely context specific is crucial: a surrogate's validity in one clinical setting cannot simply be assumed in another without appropriate supporting data. One biomarker cannot be assumed to be used for every therapy.

BIOMARKER SIGNALS FROM SECONDARY ANALYSES: HOW TO INTERPRET?

Secondary analyses of clinical trials may reveal biomarker associations with apparent clinical relevance, but their translation into practice requires a structured, evidence-based approach. Because such findings are often exploratory, interpretation should integrate biological plausibility, analytical validity, and reproducibility before any clinical inference is drawn.

First, biological coherence must be confirmed. The biomarker should align with disease pathophysiology and the intervention's mechanism of action, ideally supported by genetic, basic science, translational, or pharmacodynamic data. Analytical performance must satisfy core standards, including adequate precision in the decision range, linearity, defined

limits of detection and quantification, and reproducibility across sites and reagent lots.

Next, robustness and transportability of a candidate biomarker should be verified, ensuring consistent performance across relevant subgroups (age, sex, race and ethnicity, kidney function, adiposity), with prespecified covariate adjustments as appropriate. Clinical and statistical credibility is strengthened when both the direction and magnitude of the association are replicated in at least 2 independent data sets or trials. A pragmatic benchmark informed by large CV outcome trials and prognostic-factor work, an absolute risk difference on the order of 2 to 3 percentage points for major clinical events, or an HR of roughly 1.5 per SD change in the biomarker (or 0.67 if protective), generally reflects a moderate to strong, and therefore potentially actionable, effect size.⁴¹⁻⁴³ Gains in model calibration and net clinical benefit should take precedence over modest improvements in discrimination alone. Temporal coherence, where early biomarker changes predict subsequent outcomes, and graded associations across quantiles further support biological validity. When a surrogate role is proposed, within-trial analyses should demonstrate a proportional relationship between treatment-induced biomarker changes and differences in clinical outcomes.

Finally, study quality remains critical. Sufficient event counts, prespecified hypotheses, appropriate adjustment, and adequate controls are essential to minimize false discovery.

In sum, secondary biomarker findings should be regarded as hypothesis generating until they meet predefined standards of biological plausibility, analytical reliability, and independent confirmation. Examples of applications of this framework are provided in [Table 4](#).^{44,45}

METHODOLOGIC ASPECTS

Developing and deploying biomarkers in clinical trials require rigorous attention to assay validation, standardized sampling and handling, and robust biobanking and data systems. [Figure 2](#) summarizes key sources of variability and mitigation strategies, which are explored in the following sections. [Supplemental Table 2](#) provides a practical checklist to ensure quality across preanalytical, analytical, and postanalytical phases.

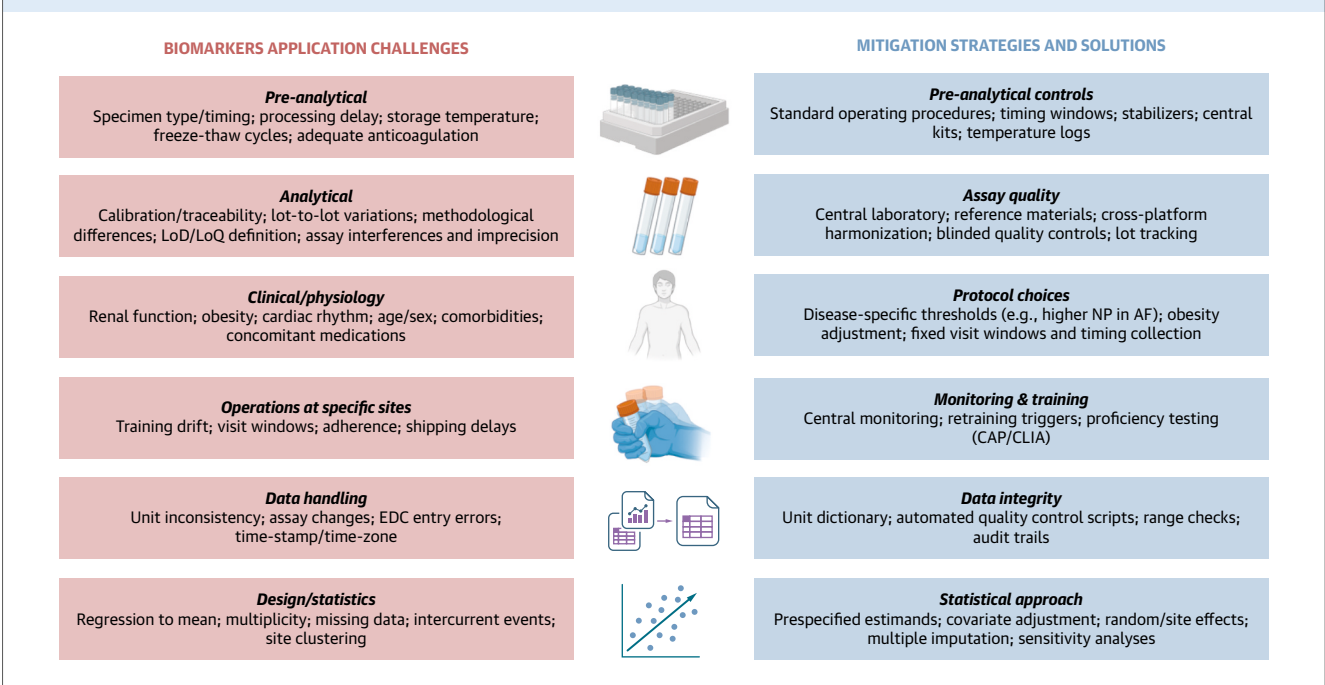
PREANALYTICAL PHASE. Biomarker values vary for reasons unrelated (or tangentially related) to the disease in question. This may include the impacts of demographics, comorbidities, concomitant drugs,

TABLE 4 Framework for Acting on Biomarker Signals From Secondary Analyses: 2 Examples

Dimension	NT-proBNP in HFREF	Lp-PLA ₂ With Darapladib ^{44,45}
Biology aligned with disease and MOA	On-pathway for myocardial stress/volume; consistent with mechanisms of HF therapies	Target inhibition did not translate to outcome benefit; biological linkage to treatment effect weak
Assay fitness (precision, linearity, LoD/LoQ, reproducibility)	Mature, standardized assays with cross-site reproducibility	Assays available, but harmonization less established; deployability undermined by absent efficacy
Transportability across key modifiers	Generally stable across age, sex, ancestry, renal function, adiposity; subgroup adjustments feasible	No persuasive, stable treatment-biomarker relationships across subgroups
Replication across independent data sets/trials	Robust replication for prognostic value and trajectory associations across trials/cohorts	Two large RCTs neutral on primary endpoints despite substantial Lp-PLA ₂ reduction; no interaction
Effect size anchors (risk difference; HR per SD)	Effects often sufficient to inform enrichment/stratification; meaningful absolute risk in high-risk groups	No clinically useful treatment-linked effect; prognostic association alone not decision-informing
Calibration and decision-curve net benefit	Acceptable calibration and positive net benefit in many contexts (baseline model dependent)	No net benefit for treatment selection given neutral outcomes
Temporality and biological gradient	Early reductions predict later outcomes; graded relationships across quantiles	Within-trial biomarker changes did not track outcome benefit
Surrogacy considerations	Trial-level surrogacy not definitive; patient-level links supportive but insufficient to replace hard outcomes	Surrogacy not supported: biomarker change uncoupled from clinical benefit
Study quality, multiplicity, bias checks	Numerous studies with adequate events; many analyses prespecified or hierarchically tested	Two well-powered RCTs with adequate precision; negative findings considered informative
Recommended action	Act with guardrails: use for enrichment, stratification, and adaptive monitoring; retain hard outcomes as primary outcomes; avoid label narrowing solely on NT-proBNP.	Do not act for selection/adaptation; avoid narrowing candidates on the basis of Lp-PLA ₂ ; if interest persists, pursue mechanistic or assay work rather than clinical rollout.

LoD = limit of detection; LoQ = limit of quantification; Lp-PLA₂ = lipoprotein-associated phospholipase A₂;MOA = mechanism of action; RCT = randomized controlled trial; TIMI = Thrombolysis In Myocardial Infarction; other abbreviations as in Tables 1 and 3.

FIGURE 2 Sources of Variability and Mitigation Across the Biomarker Life Cycle



A 2-column framework summarizing major sources of variability in multicenter biomarker trials and paired strategies such as standardized processing, assay quality programs, central laboratories, robust statistics, and data integrity safeguards to reduce noise and bias. AF = atrial fibrillation; caP = College of American Pathologists; CLIA = clinical laboratory improvement amendments; EDC = electronic data capture; LoD = limit of detection; LoQ = limit of quantification; NP = natriuretic peptide.

and genetic variation on a biomarker. For example, average NP concentrations may be lower in Black individuals (partly because of genetic factors),⁴⁶ so race-specific NP cutoffs may help promote equity in trial design.⁴⁷ NPs also vary with volume status, renal function, arrhythmias, and body mass index.⁴⁸ Sampling time, volume, and handling are critical, especially in prevention studies where absolute levels are low.⁴⁹ Moreover, some biomarkers (eg, hs-cTn) have circadian rhythm variations.⁵⁰ Harmonized collection and processing reduce variability and enable cross-study comparison. Sites should follow protocolized centrifugation and temperature control, by using ultralow-temperature freezers (−70 °C to −80 °C), and minimize freeze-thaw cycles to preserve sample integrity.

ANALYTICAL PHASE. Centralized core laboratories are preferred because they standardize measurement and reporting, enable batch testing on the same instruments and reagent lots, and participate in external quality assessment to improve interlaboratory consistency. When use of a core laboratory is not feasible, it is essential to harmonize assays, reagents, and analytical platforms across centers to maintain comparability. Early integration of these practices strengthens links between biomarker data and outcomes, particularly in large multicenter CV trials.⁵¹

POSTANALYTICAL PHASE. Results must be interpreted in the context of the study group and assay-specific reference intervals, with explicit consideration of preanalytical factors. When defining cutoffs, intraindividual and interindividual variability must be quantified so that cross-sectional values, longitudinal changes, and between-patient differences are interpreted correctly.⁵² The reference change value integrates analytical imprecision with within-person biological variation and is essential when using change-based criteria for decision making.⁵³ Assays should document sensitivity, specificity, precision, linearity, and reportable ranges, supported by routine intra-assay and inter-assay checks and quality controls; robust validation underpins dosing, monitoring, and safety detection.⁵⁴ [Supplemental Table 3](#) summarizes matrix and units, biological variation, and key caveats.

Postanalytical decisions also affect trial design and generalizability. Enrollment frequently hinges on biomarker thresholds. Enriching for high-risk profiles can increase event rates and statistical power but may exclude lower-risk patients who could still benefit. For example, in chronic kidney disease, large

outcome trials of renin-angiotensin system blockade and SGLT2 inhibitors have required substantial albuminuria (eg, UACR ≥200-300 mg/g) as an inclusion criterion.^{55,56} In these studies, the demonstrated kidney and CV benefits pertain to patients within this high-risk albuminuric range; the cutoff does not imply that benefit begins only precisely at that value, but rather it defines the risk spectrum in which efficacy has been shown and around which regulatory and reimbursement decisions tend to coalesce. If assays are imprecise or poorly standardized, patients with “borderline” values near the entry threshold may be misclassified—incorrectly included or excluded from the trial group or from real-world eligibility, thereby shifting the overall risk profile, diluting observed treatment effects, and biasing safety and efficacy estimates. This issue illustrates the importance of validated thresholds, centralized (or rigorously harmonized) measurement, and explicit consideration of measurement error when biomarker cutoffs are used to determine eligibility, enrichment, or treatment decisions.

BARRIERS TO BIOMARKER ADOPTION

Despite technical progress, most candidate biomarkers fail to influence trial design, regulatory decisions, or clinical practice. The primary obstacles are not regulatory per se, but more often they are related to limited clinical relevance and actionability and weak commercial incentives to change decisions or labels ([Table 5](#)). Even analytically robust markers rarely gain traction unless they clearly improve patient selection, guide therapy, or reshape endpoints. A visual overview of where most efforts stall in the biomarker pipeline is shown in [Figure 3](#).

CLINICAL-BIOLOGICAL GAP (ACTIONABILITY). Associations with risk do not guarantee treatment guidance and therefore surrogacy. In CV-renal-metabolic diseases, few biomarkers meet the stringent bar for surrogacy across mechanisms; well-known mismatches include albuminuria vs long-term glomerular filtration rate trajectories and NPs vs outcomes across drug classes (eg, SGLT2-inhibitors with modest NP effects yet large outcome benefits;²² conversely, large NP reductions without proportionate outcome gains in HFpEF trials⁵⁷).

EVIDENTIARY SHORTFALLS. Adoption stalls when the context of use is poorly defined and prospective evidence linking biomarker change to patient-important outcomes is sparse. Many CV biomarkers remain “supportive” only, reflecting incomplete,

TABLE 5 Barriers to Adoption of Circulating Biomarkers in Cardiovascular Drug and Device Development and Practical Solutions

Barrier	How It Manifests (Symptoms/Examples)	Why It Happens (Root Causes)	Practical Mitigations (Near-Term)	Structural Enablers (Longer-Term)	Phases Most Affected	Metrics to Track
Clinical-biological gap (actionability)	Strong risk association but weak/variable treatment-induced change; off-pathway PD signals; discordance such as limited NP change with SGLT2i despite outcome benefit, or large NP reductions without proportional benefit in some HFpEF contexts	Biomarker not on causal pathway; counter-regulation; heterogeneous disease biology; focus on prediction rather than decision impact	Declare context of use pre-FPI; require mechanistic rationale; prespecify PD targets and RCVs; triangulate with functional/imaging endpoints before dose/parameter decisions	Prospective embedding within outcomes trials; mediation and surrogate-threshold analyses across programs; shared codebooks for mechanism alignment	P2-P3 (design/efficacy interpretation)	Within-trial treatment → biomarker → outcome linkage; decision-curve net benefit at clinical thresholds
Evidentiary shortfalls	Post hoc signals; small, single-center studies; lack of replication; unclear threshold for meaningful change	Underpowered designs; fragmented datasets; absence of prespecified plans; multiplicity without control	Prospective analysis plans; hierarchical testing; external validation/temporal holdout; preregistered models and thresholds	Cross-trial biobanks; shared data platforms with harmonized metadata; funding for replication studies	P2 (signal detection) and P4 (real-world evidence)	Number of independent replications; calibration (O:E), slope/intercept; site-level transportability
Analytical and operational friction	Site-to-site variability; lot-to-lot drift; delayed turnaround that impairs adaptive rules; inconsistent matrices/units	Noncentralized testing; insufficient QC/EQA; storage/handling variability; vendor changes midtrial	Central laboratories or harmonized platforms; EQA participation; locked lots/instruments; SOPs for sampling/handling; minimize freeze-thaw; predefined RCV	Reference materials and open proficiency panels; assay-qualification networks; procurement policies that limit midtrial platform drift	P1-P3 (all adaptive/eligibility decisions)	Intersite coefficient of variation %; drift indices; turnaround time vs decision windows; nonconformance rates
Transportability and equity	Misclassification near cutoffs across age/sex/ancestry/kidney function/BMI/rhythm; reduced generalizability; subgroup disparities	Single-threshold use without partitioning or adjustment; limited representation in derivation cohorts	Partitioned cutpoints or covariate adjustment; preplanned subgroup calibration; centralized adjudication for composites	Diverse enrollment targets; standardized metadata capture; equity-focused performance reporting	P3-P4	Calibration and net benefit by subgroup; differential misclassification; representation indices
Implementation and data infrastructure	Inability to pool samples/data; incompatible formats; missing metadata; limited auditability of AI/ML pipelines	Lack of standards; IP/governance barriers; siloed biobanks	Common data models; unit/matrix standards; reproducible analysis containers; audit trails for AI/ML	Open repositories; precompetitive consortia; governance enabling secure data sharing and external validation	P2-P4	FAIRness scores; external reproduction rate; time to data share
Economics and incentives	High assay/centralization cost with unclear ROI; limited payer uptake; minimal effect on label/workflow	Biomarkers not tied to trial efficiency or approvable claims; misaligned payer/regulatory incentives	Business cases linking biomarker use to smaller/faster trials or risk reduction; include health-economic endpoints; early payer engagement	Qualification pathways with shared data; reimbursement aligned to decision-changing uses; public-private funding for replication	P2-P3	Cost per primary event; time-to-readout reduction; payer coverage decisions; incremental net benefit

Continued on the next page

TABLE 5 Continued						
Barrier	How It Manifests (Symptoms/Examples)	Why It Happens (Root Causes)	Practical Mitigations (Near-Term)	Structural Enablers (Longer-Term)	Phases Most Affected	Metrics to Track
Device-specific challenges	Operator dependence; protocol heterogeneity; lack of responsive physiological/imaging biomarkers for parameter-ranging; weak sham-controlled evidence	Learning curves; site variation; telemetry not standardized; limited central cores	Central imaging/physiology cores; standardized acquisition; sham-controlled designs where feasible; parameter-ranging frameworks tied to physiologic PD	Interoperable device data standards; postmarket registries with linked laboratories; governance for firmware/algorithm updates	P2-P4	Between-site variance; responder identification accuracy; device alert PPV/NPV; reintervention rates
Regulatory expectations gap	Surrogate claims without within-trial linkage; confusion of prognostic vs surrogate roles; late protocol revisions	Unclear context of use; insufficient early advice; conflating analytical validation with surrogacy	Early FDA/EMA advice; align to validated/reasonably-likely/supportive tiers; outcomes backstops unless validated	Consensus roadmaps (ESC roundtables, multistakeholder guidance); exemplar dossiers by indication	P2-P3	Advice-to-submission concordance; approvable endpoint rate; protocol amendment frequency
AI/ML = artificial intelligence/machine learning; BMI = body-mass index; EQA = external quality assessment; ESC = European Society of Cardiology; FAIR = findable-accessible-interoperable-reusable; FPI = first patient in; HFpEF = heart failure with preserved ejection fraction; P1, P2, P3, P4 = phase 1, phase 2, phase 3, phase 4; PPV/NPV = positive/negative predictive value; QC = quality control; RCV = reference change value; ROI = return on investment; SGLT2i = sodium-glucose cotransporter 2 inhibitor; SOP = standard operating procedure; other abbreviations as in Tables 1 and 2.						

mechanism-dependent evidence. Embedding biomarker programs within large outcome trials is often necessary but resource demanding.

ANALYTICAL AND OPERATIONAL FRICTION. Pre-analytical variability (timing, handling, physiologic state), platform differences and lot-to-lot drift impede transportability and multicenter deployment. Centralized laboratories, harmonized platforms, and external quality assessment mitigate these challenges but add cost and complexity. Sustained bandwidth for multicenter validation is frequently lacking, further slowing uptake.

TRANSPORTABILITY AND EQUITY. Biomarker distributions differ by age, sex, ethnicity, kidney function, adiposity, rhythm, and other modifiers. The applicability of sex-specific or disease-specific thresholds to define abnormalities or significant deltas is still debated for many biomarkers,⁵⁸ and in some cases, this could limit external applicability.

IMPLEMENTATION AND DATA INFRASTRUCTURE. Limited access to shared reference materials, open repositories, and precompetitive consortia slows data replication. Fragmented data between trials and subsequent secondary analysis also may limit external validation.

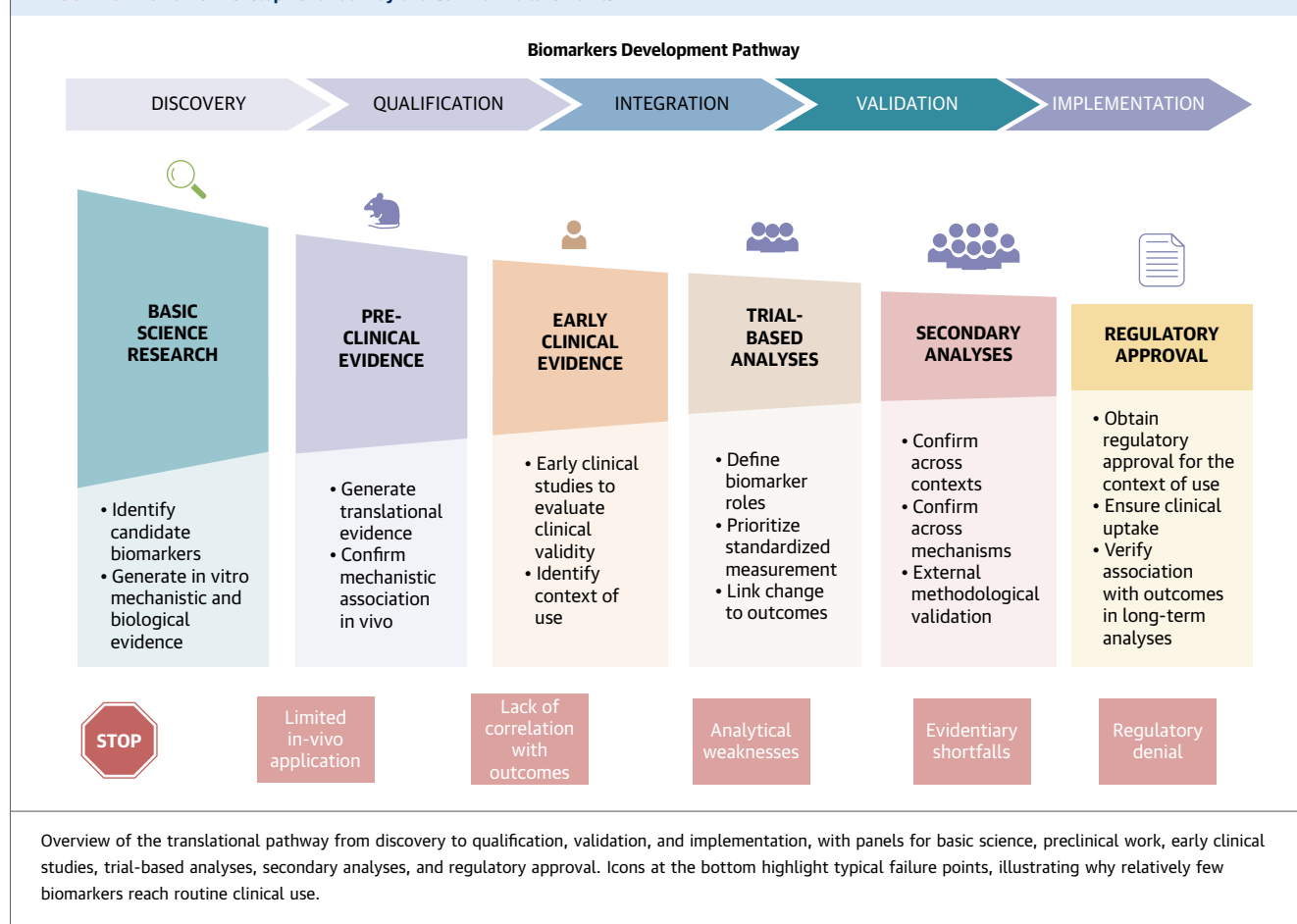
ECONOMICS AND INCENTIVES. Without clear effects on trial efficiency, labeling, or care pathways,

sponsors and health systems have little incentive to bear assay, centralization, and workflow costs at scale.

PRAGMATIC LEVERS TO OVERCOME BARRIERS. To overcome these articulated barriers, we suggest the following solutions:

1. Define precise context- of use (enrichment, pharmacodynamic or dose-ranging, safety, or surrogacy), and prespecify decision rules tied to biomarker change.
2. Embed biomarker validation prospectively within major outcome trials or registries, and create shared cross-trial biobanks.
3. Centralize and harmonize assays with external quality assessment to control drift.
4. Ensure external applicability with dedicated cut-points or reference change values, with an equity-focused design.
5. Build open data resources and precompetitive consortia to accelerate replication.
6. For devices, standardize parameter-ranging frameworks, use sham-controlled designs where feasible, and centralize imaging and physiological reads. These steps align evidentiary strength with practical incentives, thereby shifting biomarkers from descriptive measures to tools that change decisions.

FIGURE 3 Biomarker Development Pathway and Common Failure Points



EXAMPLES OF APPLICATIONS OF BIOMARKERS IN CV CLINICAL TRIALS

This section provides an overview of applications of biomarkers in clinical trials on the main forms of CV disease (Table 6, Central Illustration).

HEART FAILURE. HF is a heterogeneous syndrome, marked by diverse clinical presentations, hemodynamic profiles, and underlying mechanisms, factors that complicate the design of rigorous clinical trials. Biomarkers, particularly NPs, have become central to HF diagnosis, management, and large-scale studies.^{13,47}

In the PARADIGM-HF (A Multicenter, Randomized, Double-Blind, Parallel Group, Active-Controlled Study to Evaluate the Efficacy and Safety of LCZ696 Compared to Enalapril on Morbidity and Mortality in Patients With Chronic Heart Failure and Reduced Ejection Fraction; NCT01035255) trial, BNP ≥ 100 ng/L or NT-proBNP ≥ 600 ng/L were part of the inclusion criteria for evaluating sacubitril-valsartan against

enalapril in patients with HF with reduced ejection fraction.⁵⁹ The PARAGON-HF (Efficacy and Safety of LCZ696 Compared to Valsartan, on Morbidity and Mortality in Heart Failure Patients With Preserved Ejection Fraction; NCT01920711) trial, which investigated sacubitril-valsartan therapy in patients with HFpEF, used an NT-proBNP value of 911 ng/L (median) as the threshold for subgroup analysis.⁶⁰ The VICTORIA (A Study of Vericiguat in Participants With Heart Failure With Reduced Ejection Fraction [MK-1242-001]; NCT02861534) trial used quartiles of NT-proBNP for risk stratification in symptomatic patients with chronic HF.⁶¹ In the I-PRESERVE (Irbesartan in Heart Failure With Preserved Systolic Function; NCT00095238) trial, investigating irbesartan in HFpEF, more than 20% of patients had NT-proBNP < 125 ng/L; they had fewer comorbidities and better outcomes.⁶² Overall, these trials show how NP levels can identify homogeneous subpopulations, and increase the likelihood of detecting treatment efficacy.

TABLE 6 Applications of Laboratory Biomarkers Across Various Clinical Settings and According to Their Specific Roles

Clinical Setting	Diagnosis	Risk Prediction (Prognostic)	Pharmacodynamic/Dose-Ranging	Safety Monitoring	Surrogate Endpoints (Status)
Heart failure (HFREF/HFpEF)	NT-proBNP/BNP to support HF diagnosis and triage (age- and rhythm-aware thresholds); hs-cTn to exclude concurrent MI	NT-proBNP (baseline and serial); hs-cTn (chronic injury); GDF-15 (selected contexts)	NT-proBNP change at 4-12 wk for mechanisms with expected NP responsiveness (neurohormonal/hemodynamic); limited or delayed NP response with SGLT2i	hs-cTn for myocardial injury; creatinine/K ⁺ with RAAS/SGLT2/MRA; NT-proBNP rise with worsening congestion (phase-2 signals of likely harm reported in REALIZE-K and with volemexin in HFpEF)	None validated for HF outcomes; NT-proBNP supportive for enrichment/PD, not a qualified surrogate
Cardiovascular disease prevention/ASCVD	Lipid panel to diagnose dyslipidemia; ApoB for particle number; Lp(a) to identify inherited risk elevation	ApoB/non-HDL-C; Lp(a); hs-CRP (inflammatory risk)	LDL-C/ApoB lowering for exposure-response and dose selection across lipid-lowering classes	ALT/AST for hepatotoxicity with statins and other agents; CK for myopathy when symptomatic; glucose parameters for metabolic effects	LDL-C validated in lipid-lowering programs; ApoB/non-HDL-C supportive to program-dependent
Chronic coronary syndrome	hs-cTn (to exclude ongoing injury); lipid panel and Lp(a) to characterize etiologic risk; HbA _{1c} for diabetes context	hs-cTn (stable disease risk), NT-proBNP (ventricular stress), hs-CRP (inflammation)	LDL-C/ApoB response to therapy; platelet reactivity or P2Y ₁₂ levels in selected mechanistic studies (exploratory)	Hemoglobin/platelets for bleeding risk with antithrombotic agents; liver/kidney panels for polypharmacy	No qualified laboratory surrogate for ischemic outcomes in CCS; lipid metrics supportive unless in lipid-lowering efficacy claims
Cardiomyopathies and rare diseases	Disease-specific markers: lyso-Gb3 (Fabry); genetic testing; NT-proBNP/hs-cTn for cardiac involvement	NT-proBNP and hs-cTn for amyloidosis and other infiltrative CM; eGFR/UACR for cardiorenal involvement; sST2 in selected contexts	Lyso-Gb3 change with Fabry-directed therapy; NT-proBNP as PD in amyloidosis (context dependent)	Organ injury panels aligned to therapy (liver/kidney); hs-cTn for myocardial injury during therapy initiation/switch	No broadly validated laboratory surrogates; lyso-Gb3 supportive in Fabry; NT-proBNP supportive, mechanism-dependent
Atrial fibrillation	No definitive circulating diagnostic; NT-proBNP may support atrial disease burden; D-dimer used in specific contexts	NT-proBNP, hs-cTn, GDF-15 within ABC-Stroke/ABC-Bleeding frameworks alongside clinical scores	Limited laboratory PD markers; exploratory use of NP dynamics around rhythm-control strategies in research settings	Renal function and anti-Xa (where indicated) for anticoagulation management; hemoglobin for occult bleeding	No qualified laboratory surrogate for AF hard outcomes; biomarkers supportive for enrichment/risk only

ABC = age-biomarkers-clinical; AF = atrial fibrillation; AI = artificial intelligence; ApoB = apolipoprotein B; BNP = B-type natriuretic peptide; CCS = chronic coronary syndrome; CK = creatine kinase; CM = cardiomyopathy; GDF-15 = growth differentiation factor-15; HFpEF = HF with preserved ejection fraction; IL-6 = interleukin-6; K⁺ = potassium; MI = myocardial infarction; MRA = mineralocorticoid receptor antagonist; RAAS = renin-angiotensin-aldosterone system; REALIZE-K = Study to Assess Efficacy and Safety of Sodium Zirconium Cyclosilicate for the Management of High Potassium in Patients With Symptomatic HFREF Receiving Spironolactone; sST2 = soluble ST2; other abbreviations as on [Tables 1 to 3 and 5](#).

NPs have also been explored as potential surrogate endpoints. Post hoc analyses of landmark trials, including PARADIGM-HF,¹⁴ DAPA-HF,⁶³ and VICTORIA,⁶⁴ have shown that early treatment-induced reductions in B-type natriuretic peptide (BNP) or NT-proBNP (typically $\geq 20\%$ -30%) are strongly associated with subsequent decreases in CV death or HF hospitalization, independent of baseline risk factors. Mechanistic studies such as PROVE-HF (Effects of Sacubitril/Valsartan Therapy on Biomarkers, Myocardial Remodeling, and Outcomes; [NCT02887183](#))¹⁵ and EVALUATE-HF (Study of Effects of Sacubitril/Valsartan vs Enalapril on Aortic Stiffness in Patients With Mild to Moderate HF With Reduced Ejection Fraction; [NCT02874794](#))¹⁶ further support this concept: proportional declines in NT-proBNP closely paralleled improvements in left ventricular volumes and Kansas City Cardiomyopathy Questionnaire scores, thus underscoring biological plausibility for NPs as integrative indicators of

ventricular wall stress and reverse remodeling. From a regulatory perspective, both the FDA⁶⁵ and the European Medicines Agency (EMA)⁶⁶ consider short-term changes in NPs to be “reasonably likely” surrogate biomarkers that may justify accelerated approval pathways, provided confirmatory outcome trials are underway. Pooled data initiatives and prospective biomarker-endpoint trials are warranted to clarify the quantitative relationship between NT-proBNP trajectories and clinical benefit, potentially paving the way for their full qualification as surrogate endpoints.

Beyond NPs, multiomics technologies offer deeper insights into HF pathophysiology and the variability among patient subgroups.^{67,68} One example is the prospective, open label HOMAGE (Heart ‘Omics’ in AGEing) trial,⁶⁹ where older adults at elevated risk of HF were randomized to spironolactone or placebo. The analyses indicated that spironolactone modulates multiple systems, including the renin-angiotensin-

CENTRAL ILLUSTRATION Circulating Biomarkers Across Cardiovascular and Cardiorenal Indications

 LEVEL OF EVIDENCE	 HEART FAILURE	 CV PREVENTION	 CCS	 ATRIAL FIBRILLATION	 CARDIOMYOPATHY / RARE DISEASES	 KIDNEY DISEASE	 METABOLIC SYNDROME
 DIAGNOSTIC	NT-proBNP Hs-cTn	Lipid panel Lp(a), ApoB NT-proBNP Hs-cTn, UACR Hs-CRP Multiomics	Hs-cTn	NT-proBNP Hs-cTn	NT-proBNP hs-cTn lyso-Gb3 genetic testing microRNAs	Creatinine eGFR UACR Cistatin-C	Hb1Ac 2h-OGTT
 ENRICHING-RISK STRATIFICATION	NT-proBNP Hs-cTn sST2, Gal3, GDF15, Multiomics	NT-proBNP Hs-cTn ApoB/non-HDL-C Lp(a) hs-CRP Multiomics	Hs-cTn NT-proBNP ApoB/non-HDL-C Lp(a) hs-CRP	NT-proBNP hs-cTn GDF-15	NT-proBNP,hs-cTn, eGFR/UACR sST2, Gal3, GDF15, lyso-Gb3	Creatinine eGFR UACR	HbA1c microalbuminuria
 SAFETY	NT-proBNP Hs-cTn Ions Creatinine, eGFR AST, ALT	AST, ALT CK	Blood count Creatinine, eGFR AST, ALT	Blood count Creatinine, eGFR AST, ALT	Blood count Creatinine, eGFR AST, ALT NTproBNP, hs-cTn	creatinine, BUN NGAL, cysC PFC Index (CLU, CysC, KIM-1, NAG, NGAL, OPN)	Glucose monitoring, AST, ALT,
 SURROGATE	Not validated	LDL-C ApoB/non-HDL-C, hs-CRP	Not validated Hs-cTn, hs-CRP	Not validated	Not validated NT-proBNP for AL amyloidosis	eGFR slope UACR	Hb1Ac LDL-C

Aimo A, et al. JACC. 2026;87(15):1950-1972.

Matrix showing major circulating biomarkers mapped by level of evidence and clinical setting. Within each cell, representative biomarkers are displayed in the roles for which they are currently supported. (Icon images from BioRender.) AL = light chain; ALT = alanine aminotransferase; ApoB = apolipoprotein B; AST = aspartate aminotransferase; BUN = blood urea nitrogen; CCS = chronic coronary syndrome; CK = creatine kinase; CKD = chronic kidney disease; CLU = clusterin; CV = cardiovascular; cysC = cystatin-C; eGFR = estimated glomerular filtration rate; Gal-3 = galectin-3; GDF-15 = growth differentiation factor-15; HbA_{1c} = glycated hemoglobin; HDL-C = high-density lipoprotein cholesterol; HF = heart failure; hs-CRP = high-sensitivity C-reactive protein; hs-cTn = high-sensitivity cardiac troponin; KIM-1 = kidney injury molecule-1; LDL-C = low-density lipoprotein cholesterol; Lp(a) = lipoprotein (a); lyso-Gb3 = lysoglobotriaosylceramide; NAG = N-acetyl-beta-d-glucosaminidase; NGAL = neutrophil gelatinase-associated lipocalin; NP = natriuretic peptide; NT-proBNP = N-terminal pro-B-type natriuretic peptide; OGTT = oral glucose tolerance test; OPN = osteopontin; sST2 = soluble suppression of tumorigenesis 2; PFC = PSTC/FNIH Kidney Safety Biomarker Qualification; UACR = urine albumin-to-creatinine ratio.

aldosterone pathway, inflammatory signaling, adipokine networks, and extracellular matrix metabolism.⁶⁹ In particular, patients receiving spironolactone had greater reductions in procollagen type I C-terminal propeptide (PICP) (mean difference: -8.1; 95% CI: -11.9 to -4.3 µg/L; $P < 0.001$),⁶⁹ and PICP/collagen type 1 C-terminal telopeptide ratio (mean difference: -2.9; 95% CI: -4.3 to -1.5; $P < 0.001$).⁶⁹ Tracking these shifts in protein expression can help researchers confirm target engagement and optimize dosing decisions in early-phase clinical studies.

Another example of how biomarkers may inform drug targets was shown in the BIOSTAT-CHF (Systems BIOlogy Study to Tailored Treatment in Chronic Heart Failure) study.⁷⁰ Machine learning identified pathways related to disease progression and early mortality.⁷⁰ The strongest pathways were related to the epidermal growth factor receptors family (ERBB)

and their downstream effects on the phosphoinositide 3-kinase (PI3K) and mitogen-activated protein kinase (MAPK) pathways; this is relevant given an ongoing study on a therapy affecting the ERB pathway (Study of JK07 in Patients With Chronic Heart Failure [RENEU-HF]; [NCT06369298](#)).

In summary, biomarkers (particularly NPs) play a pivotal role in HF trials: they refine patient selection, uncover early efficacy signals, and illuminate key disease mechanisms. Nonetheless, NPs still serve mainly as “bridge” endpoints, and clear, consensus standards are needed before BNP or NT-proBNP can be accepted as fully approvable endpoints. Furthermore, the dominance of BNP and NT-proBNP in HF trials may arguably have slowed development of newer biomarkers that may play different roles in HF trials. As multiomics technologies continue to mature, integrating these data with existing

biomarkers promises to streamline trial design even further and accelerate progress toward truly personalized, more effective HF therapies.

CARDIOVASCULAR DISEASE PREVENTION. Biomarkers have emerged as powerful tools for the early detection and prevention of CV disease.⁷¹⁻⁷³ The noninvasiveness of their use enables repeated measurements over time, offering a dynamic view of cardiac physiology and revealing subtle changes that may precede clinical symptoms.

A growing body of evidence supports the use of circulating biomarkers to identify individuals with asymptomatic cardiac disease who may benefit from therapeutic interventions. For example, modest elevations in troponin I in patients without a previous diagnosis of HF or coronary artery disease predict a higher risk of future CV events. In the ARIC (Atherosclerosis Risk In Communities) study,⁷⁴ including 8,121 patients, participants with hs-TnI ≥ 3.8 ng/L (value still below the 99th upper reference level) had greater incident coronary heart disease (adjusted HR: 2.20; 95% CI: 1.64-2.95), HF hospitalization (adjusted HR: 4.20; 95% CI: 3.28-5.37), and all-cause mortality (adjusted HR: 1.83; 95% CI: 1.56-2.14). In 5,324 community-dwelling older adults in the same cohort, a single measurement of hs-cTnT or NT-proBNP identified stage B HF risk just as well as a comprehensive echocardiogram (~ 2 -fold risk elevation vs low-risk stage A). When both biomarkers and echocardiographic findings were abnormal, the risk was synergistically higher, more than 5-fold for incident HF and ~ 3 -fold for all-cause death.⁷⁵

In the MESA (Multi-Ethnic Study of Atherosclerosis) cohort,⁷⁶ hs-cTnT was able to refine risk prediction even on top of the coronary calcium score, with a risk conferred by hs-cTnT lower than the limit of detection and a coronary artery calcium score >0 of 11.1 events per 1,000 person-years (HR: 2.74; 95% CI: 1.96-3.83; $P < 0.001$) and by hs-cTnT greater than or equal to the limit of detection and a calcium score >0 of 22.6 per 1,000 person-years (HR: 3.50; 95% CI: 2.60-4.70; $P < 0.001$).⁷⁶ Again, in the MESA cohort,⁷⁷ increased NT-proBNP levels were associated with early left ventricular dysfunction, a finding suggesting that these biomarkers can detect myocardial stress or injury at a subclinical stage. In the SPRINT (Systolic Blood Pressure Intervention Trial) cohort, baseline (HR: 1.23, 95% CI: 1.08-1.39, per 1-unit increase in log[hs-cTnI]) and increased hs-cTnI at 1 year were independently associated with CV risk, and the treatment effect of intensive blood pressure lowering was greater in individuals with relatively higher hs-cTnI at baseline (≥ 6 ng/L in

men, ≥ 4 ng/L in women). Interestingly, hs-cTnI and hs-cTnT appeared to provide complementary risk information, thus underlying that even different isoforms of a laboratory biomarker may deploy different shades of risk assessment.

CV risk assessment in high-risk groups is another important topic in disease prevention. In persons with diabetes mellitus who were treated with canagliflozin or placebo in the CANVAS (Canagliflozin Cardiovascular Assessment Study; [NCT01032629](#)) trial,¹⁷ almost 40% had NT-proBNP levels higher than 125 ng/L with a reduction in NT-proBNP of 11% after 1 year (geometric mean ratio for canagliflozin vs placebo = 0.89; 95% CI: 0.84-0.94; $P < 0.001$). Approximately 10% of the effects of canagliflozin on HF hospitalization were reflected in NT-proBNP lowering, underlying that the benefits of this drug are related also to different pathways.¹⁷ The powerful link between NP concentrations and risk in persons with diabetes has led to the launch of prospective HF prevention trials that prominently feature elevated NP concentrations as inclusion criteria among individuals with cardiometabolic disease (Phase III Study Investigating Heart Failure and Cardiovascular Death With Baxdrostat in Combination With Dapagliflozin [Prevent-HF]; [NCT06677060](#); and A Multicenter, International, Randomized, Double-blind, Placebo-controlled Clinical Trial of the Aldosterone Synthase Inhibitor BI 690517 in Combination With Empagliflozin in Patients With Chronic Kidney Disease; [NCT06531824](#)).

Besides cardiac biomarkers, UACR is emerging as a potential marker to recognize the risk of HF in diabetic nephropathy. In the FIGARO-DKD (Efficacy and Safety of Finerenone in Subjects With Type 2 Diabetes Mellitus and the Clinical Diagnosis of Diabetic Kidney Disease; [NCT02545049](#)) trial,¹⁸ including patients with diabetic kidney disease of various degrees and a UACR >30 mg/g, finerenone therapy improved CV outcomes, particularly in terms of reduction in HF hospitalization.¹⁸ This association indicates that moderate or severely increased albuminuria potentially serves as a useful marker to forecast HF in high-risk patients.¹⁹

Circulating biomarkers can also serve as surrogate endpoints by providing early signals of therapeutic efficacy. In the JUPITER (Crestor 20 mg Versus Placebo in Prevention of Cardiovascular Events; [NCT00239681](#)) study, participants allocated to rosuvastatin who achieved LDL-C levels <1.8 mmol/L had a 55% reduction in CV events than patients receiving placebo (HR: 0.45; 95% CI: 0.34-0.60; $P < 0.001$), and those achieving hs-C-reactive protein (hs-CRP) less than 2 mg/L had a 62% reduction (HR: 0.38; 95% CI:

0.26-0.56; $P < 0.001$).⁷² This finding reinforces the notion that early biomarker changes can forecast long-term benefits, therefore acting as potentially useful and precise surrogate endpoints.

The use of circulating biomarkers in prevention studies is not without challenges. Participants with preclinical disease often have lower concentrations of risk biomarkers; given that test accuracy typically declines when biomarker concentrations are low, this could affect fidelity of the results. For example, very low hs-cTn concentrations may have lower precision.⁷⁸ It is essential to prespecify both the threshold that constitutes a meaningful change over time and the sampling intervals at which the biomarker will be measured. Looking ahead, the integration of laboratory biomarkers with advanced multiomics platforms promises to refine CV disease prevention strategies further. Emerging analytical techniques, such as machine learning, are poised to identify novel biomarker patterns that predict the transition from subclinical dysfunction to overt disease. To date, measurement of CV biomarkers as a general risk marker in all unselected individuals cannot be recommended,⁷³ but selected measurement is promising in intermediate-risk patients, particularly for refining CV trial enrollment and outcomes definitions.

CHRONIC CORONARY SYNDROME. Chronic coronary syndrome encompasses a range of manifestations, from stable angina to asymptomatic ischemia. Circulating biomarkers such as hs-cTn, C-reactive protein, and NPs correlate with underlying myocardial stress and systemic inflammation. In the CANTOS (Cardiovascular Risk Reduction Study [Reduction in Recurrent Major CV Disease Events]; [NCT01327846](#)) trial,⁷⁹ for example, treatment with canakinumab led to significant reductions in hs-CRP levels (up to 41% in patients treated with 300 mg compared with placebo), and this finding correlated with decreased CV atherosclerotic event rates (HR: 0.86; 95% CI: 0.75-0.99; $P = 0.031$ in the 300-mg group),⁷⁹ thus underscoring the potential of this biomarker as a surrogate endpoint in this specific treatment pathway.

Baseline biomarker values can aid in risk stratification and treatment decision making in chronic coronary syndrome. For instance, in the BARI2D (Bypass Angioplasty Revascularization Investigation in Type 2 Diabetes; [NCT00006305](#)) trial,⁸⁰ hs-cTnT concentration was an independent predictor of adverse outcomes in patients with diabetes and ischemic heart disease (HR: 1.85; 95% CI: 1.48-2.32; $P < 0.001$ for hs-cTnT >14 ng/L), and elevated values did not identify those patients who benefited from

prompt coronary revascularization compared with medical therapy alone (HR: 0.96; 95% CI: 0.74-1.25, for patients with hs-cTnT >14 ng/L).

The setting of chronic coronary syndrome also highlights the need for clear and standardized definitions. A relevant example comes from trials on coronary revascularization in the setting of left main coronary artery disease, such as the EXCEL (Evaluation of XIENCE Versus Coronary Artery Bypass Surgery for Effectiveness of Left Main Revascularization; [NCT01205776](#))⁸¹ and NOBLE (Nordic-Baltic-British Left Main Revascularization Study; [NCT01496651](#))⁸² trials. EXCEL included periprocedural myocardial infarction as part of the endpoint, whereas NOBLE did not consider biomarkers or periprocedural infarction as part of the primary endpoint, thereby possibly contributing to the different conclusions. Hence definitions that incorporate laboratory biomarkers and their consistent use across clinical trials contribute to the comparability of results and their clinical robustness.⁸³

OTHER CLINICAL SCENARIOS. Biomarkers may provide particular value in trials of rare diseases, where patient enrollment is difficult and achieving hard clinical endpoints is challenging. In light-chain amyloidosis, for example, rising NT-proBNP levels stem not only from hemodynamic stress but also from the direct cardiotoxicity of circulating light chains, creating a clear mechanistic link between the marker and the disease process.⁸⁴ Extensive retrospective and prospective evidence supports NT-proBNP as a clinical trial endpoint, placing it among prioritized outcomes alongside mortality, hospitalizations, and the 6-minute walk test.⁸⁵ Novel biomarkers, including specific microRNAs and exosomal proteins, are being investigated as potential indicators of disease mechanisms and treatment response. For example, a novel circulating noncoding small RNA has been identified for the diagnosis of myocarditis⁸⁶ (area under the receiver-operating-characteristic curve for the novel RNA hsa-Chr8:96 for distinguishing patients with acute myocarditis from those with myocardial infarction was 0.927; 95% CI: 0.879-0.975), thus potentially enhancing disease detection and homogenizing patients' selection for trials. Disease-specific endpoints and biomarker roles are detailed in [Supplemental Table 4](#).

Trials of more common diseases such as atrial fibrillation can be enhanced by the application of biomarkers. For early detection, elevated concentrations of NT-proBNP predict the development of atrial fibrillation in several large cohort studies,

including the Framingham Heart Study⁸⁷ (during a follow-up of 12 years, HR for atrial fibrillation per 1-SD increment of rank-normalized NT-proBNP: 1.73; 95% CI: 1.52-1.96). The hs-cTnT concentrations, even when modestly increased, were associated with incident AF over and above traditional risk factors; indeed, in the Cardiovascular Health Study cohort, baseline hs-cTnT was associated with incident AF up to 24.53 ng/L (HR: 1.67 per log unit hs-cTnT; 95% CI: 1.52-1.83), with no additional increased risk above 24.53 ng/L.⁸⁸ In the PREDICT-AF (PREventive Left Atrial appenDage Resection for predICtion of fuTure Atrial Fibrillation; [NCT03130985](#)) trial, several biomarkers related to extracellular matrix and atrial remodeling were associated with future atrial fibrillation development.⁸⁹ Therefore, biomarkers can be incorporated into patient selection criteria to enrich study cohorts with individuals at higher risk of atrial fibrillation or exhibiting early signs of atrial dysfunction.

Biomarkers can guide dedicated subanalyses and identify patients at higher risk of adverse outcomes in atrial fibrillation trials. In ARISTOTLE (Apixaban for Reduction in Stroke and other Thromboembolic Events in Atrial Fibrillation; [NCT00412984](#)),⁹⁰ patients with higher levels of hs-cTnT and hs-cTnI had an increased risk of adverse outcomes (adjusted HR for cardiac death for patients in the highest quartile: 4.31 95% CI: 2.91-6.37 for hs-cTnT and 4.52; 95% CI: 3.05-6.70 for hs-cTnI); dedicated subanalyses showed how the benefits of apixaban were consistent in these higher-risk subsets.^{90,91}

REGULATORY FRAMEWORK

Regulators evaluate circulating biomarkers along 2 converging evidence lanes—analytical readiness and clinical or statistical validation—that culminate in a decision on the biomarker's regulatory *context of use* (eg, enrichment tool, pharmacodynamic/dose-ranging marker, or surrogate endpoint). Acceptance of a biomarker hinges on robust assay performance, pathway biology for the intended mechanism, and evidence that treatment-induced changes in the biomarker track with patient-important outcomes in the specific setting.

In the United States, the FDA differentiates between validated surrogate endpoints—biomarkers shown across multiple studies to predict clinical benefit and to be suitable for traditional approval—and “reasonably likely” surrogate endpoints, which may support accelerated approval in serious conditions but require confirmatory outcome trials.

Biomarkers that are purely prognostic or pharmacodynamic, without evidence that their modification predicts outcomes for the mechanism and indication at hand, are generally regarded as supportive and should not carry primary efficacy claims.^{36,92}

The EMA operates a parallel structure through its qualification of novel methodologies procedure.⁹³ This pathway provides the following: 1) qualification advice on future studies; 2) qualification opinions that endorse a biomarker or method for a specific, predefined context of use; and 3) letters of support when the methodology appears promising but the data are not yet sufficient for a qualification opinion.⁹³ Each of these outputs is explicitly tied to a defined context of use, mirroring the FDA's emphasis on context-specific biomarker qualification.^{36,92}

In phase 2 trials, FDA and EMA are broadly aligned that circulating biomarkers are most useful for the following purposes: 1) confirming biological mechanism and target engagement; 2) characterizing pharmacodynamic and dose-response relationships; 3) enriching for patients at higher baseline risk or greater likelihood of benefit; and 4) early safety monitoring, provided assays are centralized or rigorously harmonized and decision thresholds are prespecified.^{36,92,93} Use of a circulating biomarker as the primary efficacy endpoint in phase 2 is typically exploratory, unless a credible surrogacy case already exists for the mechanism and disease. In HF, short-term NT-proBNP change is generally viewed as useful for decision making in enrichment and pharmacodynamic assessment, but it has not been formally qualified as a surrogate endpoint for regulatory purposes. By contrast, for LDL-C lowering in atherosclerotic CV disease, extensive randomized outcome data have led regulators to treat LDL-C as a well-established surrogate endpoint that can support efficacy claims for certain lipid-lowering drug classes when aligned with the existing evidence base. Biomarkers of inflammation (eg, hs-CRP) and safety (eg, hs-cTn for cardiotoxicity monitoring) are typically considered decision supportive rather than stand-alone efficacy endpoints.^{36,92,93}

In phase 3, regulators generally expect demonstration of benefit on clinical outcomes (eg, mortality, major adverse CV events, HF hospitalization) or, when adequately justified, on validated surrogate endpoints for traditional approval. Use of a “reasonably likely” surrogate is reserved for accelerated approval pathways in serious conditions and must be backed by an active, adequately powered outcomes program designed to confirm clinical benefit. For HF, this translates into using NT-proBNP primarily for

enrichment and pharmacodynamic readouts while retaining hard outcomes as primary endpoints. For atherosclerotic CV disease risk reduction, LDL-C lowering can anchor claims when it occurs on top of an established class-specific outcome evidence base. In both FDA and EMA settings, markers such as hs-CRP and hs-cTn are accepted components of risk stratification and safety or risk-management plans, but they do not by themselves justify efficacy claims.^{36,92,93}

Across development stages, regulators look for a coherent package: analytically robust assays; prespecified thresholds and sampling windows; within-trial analyses linking treatment-induced biomarker changes to clinical endpoints; and, where surrogacy is claimed, evidence that achieving a given biomarker effect size predicts benefit at both the trial-arm and individual-patient levels. Overreliance on surrogate claims without such linkage risks approvals that fail to improve outcomes. Conversely, strong use of biomarkers for pharmacodynamic assessment, enrichment, and safety can substantially increase phase 2 to 3 trial efficiency without requiring that the biomarker itself be qualified as a surrogate endpoint.^{36,92,93}

FROM PLAUSIBILITY TO PRACTICE: WHAT WILL IT TAKE FOR BIOMARKERS TO MATTER?

Despite their appeal, only a few circulating biomarkers currently satisfy the stringent requirements to serve as surrogate endpoints in CV-kidney-metabolic drug development. Clinical experience shows repeated mismatches between candidate biomarker changes and patient benefit. Prognostic power and biological plausibility, although necessary, are not sufficient to ensure that treatment-induced changes in a marker predict clinical benefit. The central implication is straightforward: large, well-designed outcome trials remain indispensable, and biomarker discovery and validation should be embedded within them rather than pursued in parallel.

In trial phases 1 and 2, biomarkers are most useful as decision tools—for dose-ranging, mechanistic confirmation, enrichment, and early safety—rather than as substitutes for clinical outcomes. Sponsors should define the biomarker's context of use initially (diagnostic, prognostic, pharmacodynamic or dose-ranging, safety, or putative surrogate), specify reference change values and decision thresholds, and prespecify adaptive rules for adding, dropping, or graduating doses when biomarker trajectories

support or contradict the expected drug effect. Early safety surveillance should rely on well-characterized safety biomarkers with prespecified adjudication plans, and centralized (or rigorously harmonized) measurements should be preferred to minimize noise in small, early-stage cohorts.

In phase 3, biomarkers should primarily make trials more efficient and more informative; lacking clearly defined surrogate roles, biomarkers cannot currently replace hard outcomes. Key uses include the following: 1) enriching for higher-risk or mechanism-relevant groups; 2) increasing event rates and statistical power; and 3) supporting composite or hierarchical endpoints with mechanistic insight. Clinical events (eg, mortality, major adverse CV events, HF hospitalization, kidney failure) should remain the gatekeepers for efficacy claims. In kidney indications, sustained estimated glomerular filtration rate slope and clinical kidney outcomes should anchor development, with albuminuria treated as supportive and mechanism specific. In HF, NPs are best positioned for enrichment and pharmacodynamic readouts, rather than as stand-alone surrogates for regulatory approval.

In phase 4, longitudinal biomarker monitoring linked to registries and real-world data can verify durability of benefit, detect emerging harm, and identify patient subsets whose biomarker-outcome coupling is strongest. Criteria should be prespecified for when consistent improvements in biomarker-anchored intermediate outcomes warrant formal studies aimed at label expansion, new indications, or refined dosing strategies.

Across all phases, credible biomarker evidence comes from prospective, preplanned integration into major outcome trials. Programs should include the following: 1) relate treatment-induced biomarker changes to clinical endpoints at both patient and treatment-arm levels; 2) assess calibration, discrimination, and decision-curve net benefit at plausible intervention thresholds; and 3) when surrogacy is contemplated, test for reproducible “threshold effects” where achieving a given biomarker change consistently predicts clinical benefit across trials and data sets. Operationally, assays and platforms should be locked or tightly controlled, with external quality assessments to monitor drift. Transportability analyses should be prespecified across age, sex, race and ethnicity, kidney function, adiposity, rhythm, and comorbidity, using partitioned cutpoints or covariate adjustment to avoid systematic misclassification. Importantly, trial documents should state exactly how biomarker-based rules influence dose or

parameter selection, study flow, or go/no-go decisions; if they do not change actions, the marker is descriptive rather than decision making.

The main barriers to biomarker adoption are increasingly practical rather than regulatory: limited clinical actionability, analytical and operational friction, fragmented infrastructure, and weak economic incentives. Inertia in the in vitro diagnostics industry has led to stalling of innovation in new biomarker development, whereas therapeutic development is slowed because of lack of qualified surrogates and/or fear of failing to secure regulatory approval on the basis of dependency of newer biomarkers for efficacy readouts.

To align incentives, stakeholders should do the following: 1) collaborate to develop newer biomarkers that truly advance the field, rather than depending repeatedly on the same group of markers again and again; 2) standardize assays, reporting units, and sampling matrices; 3) build shared data platforms and cross-trial biobanks with harmonized protocols and metadata; 4) preregister biomarker analysis plans and make executable scripts shareable; and 5) in device and neuromodulation programs, use sham controls and centralized imaging or physiology cores to preserve signal-to-noise ratios.

A forward-looking but pragmatic compact is therefore within reach. Reports from the European Society of Cardiology Cardiovascular Round Table outline structured pathways for biomarker development, validation, and implementation, they emphasize standardized analytical performance criteria, clear clinical use cases, and multi-stakeholder governance across academia, industry, patient organizations, and regulators.⁹⁴ Complementary proposals call for systematic biospecimen archiving, assay harmonization, and biomarker-informed adaptive designs to accelerate CV trials without compromising safety or rigor.^{3,94} Building on these frameworks, we envision precompetitive consortia that maintain open access repositories of biospecimens and curated data, support multiomic discovery pipelines, and provide shared, artificial intelligence-enabled analytics that are reproducible, bias aware, and clinically interpretable.^{3,94} In such an ecosystem, discovery, validation, and

implementation would be explicitly organized around patient benefit and decision-making impact: biomarkers would be chosen and qualified because they change what clinicians do, for whom, and when. If biomarker validation is routinely embedded within outcome trials and linked to interoperable infrastructures, circulating biomarkers can evolve from tools that merely predict risk to tools that reliably guide decisions, accelerating CV-kidney-metabolic drug development while improving equity, safety, and clinical outcomes.

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REFERENCES

1. Halim SA, Newby LK, Ohman EM. Biomarkers in cardiovascular clinical trials: past, present, future. *Clin Chem*. 2012;58:45-53.
2. Bayes-Genis A, Aimo A, Jhund P, et al. Biomarkers in heart failure clinical trials. A review from the Biomarkers Working Group of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail*. 2022;24:1767-1777.

3. Januzzi JL Jr, Canty JM, Das S, et al. Gaining efficiency in clinical trials with cardiac biomarkers: JACC review topic of the week. *J Am Coll Cardiol*. 2021;77:1922–1933.
4. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *J Am Coll Cardiol*. 2018;72:2231–2264.
5. Bozkurt B, Coats AJS, Tsutsui H, et al. Universal definition and classification of heart failure: a report of the Heart Failure Society of America, Heart Failure Association of the European Society of Cardiology, Japanese Heart Failure Society and Writing Committee of the Universal Definition of Heart Failure. *Eur J Heart Fail*. 2021;23:352–380.
6. 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 suppl 1):S27–S49.
7. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2024;105(4 suppl):S117–S314.
8. McMurray JJV, Solomon SD, Inzucchi SE, et al. Dapagliflozin in patients with heart failure and reduced ejection fraction. *N Engl J Med*. 2019;381:1995–2008.
9. Mebazaa A, Davison B, Chioncel O, et al. Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure (STRONG-HF): a multinational, open-label, randomised, trial. *Lancet*. 2022;400:1938–1952.
10. Cannon CP, Blazing MA, Giugliano RP, et al. Ezetimibe added to statin therapy after acute coronary syndromes. *N Engl J Med*. 2015;372:2387–2397.
11. Sabatine MS, Giugliano RP, Keech AC, et al. Evolocumab and clinical outcomes in patients with cardiovascular disease. *N Engl J Med*. 2017;376:1713–1722.
12. Burlina A, Brand E, Hughes D, et al. An expert consensus on the recommendations for the use of biomarkers in Fabry disease. *Mol Genet Metab*. 2023;139:107585.
13. Vaduganathan M, Claggett B, Packer M, et al. Natriuretic peptides as biomarkers of treatment response in clinical trials of heart failure. *JACC Heart Failure*. 2018;6:564–569.
14. Zile MR, Claggett BL, Prescott MF, et al. Prognostic implications of changes in N-terminal pro-B-type natriuretic peptide in patients with heart failure. *J Am Coll Cardiol*. 2016;68:2425–2436.
15. Januzzi JL Jr, Prescott MF, Butler J, et al. Association of change in N-terminal pro-B-type natriuretic peptide following initiation of sacubitril-valsartan treatment with cardiac structure and function in patients with heart failure with reduced ejection fraction. *JAMA*. 2019;322:1085–1095.
16. Myhre PL, Claggett BL, Shah AM, et al. Changes in cardiac biomarkers in association with alterations in cardiac structure and function, and health status in heart failure with reduced ejection fraction: the EVALUATE-HF trial. *Eur J Heart Fail*. 2022;24:1200–1208.
17. Januzzi JL Jr, Xu J, Li J, et al. Effects of canagliflozin on amino-terminal pro-B-type natriuretic peptide: implications for cardiovascular risk reduction. *J Am Coll Cardiol*. 2020;76:2076–2085.
18. Pitt B, Filippatos G, Agarwal R, et al. Cardiovascular events with finerenone in kidney disease and type 2 diabetes. *N Engl J Med*. 2021;385:2252–2263.
19. Khan MS, Shahid I, Anker SD, et al. Albuminuria and heart failure: JACC state-of-the-art review. *J Am Coll Cardiol*. 2023;81:270–282.
20. Li P, Taylor JMG, Kong S, Jolly S, Schipper MJ. A utility approach to individualized optimal dose selection using biomarkers. *Biom J*. 2020;62:386–397.
21. Bhatt DL, Mehta C. Adaptive designs for clinical trials. *N Engl J Med*. 2016;375:65–74.
22. Ibrahim NE, Januzzi JL. Sodium-glucose co-transporter 2 inhibitors and insights from biomarker measurement in heart failure patients. *Clin Chem*. 2021;67:79–86.
23. Dockery F, Bulpitt CJ, Agarwal S, et al. Anti-androgens increase N-terminal pro-BNP levels in men with prostate cancer. *Clin Endocrinol (Oxf)*. 2008;68:59–65.
24. Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med*. 2015;373:2117–2128.
25. Wiviott SD, Raz I, Bonaca MP, et al. Dapagliflozin and cardiovascular outcomes in type 2 diabetes. *N Engl J Med*. 2019;380:347–357.
26. Zhao X, Modur V, Carayannopoulos LN, Laterza OF. Biomarkers in pharmaceutical research. *Clin Chem*. 2015;61:1343–1353.
27. Katsurada K, Ichida M, Sakuragi M, Takehara M, Hozumi Y, Kario K. High-sensitivity troponin T as a marker to predict cardiotoxicity in breast cancer patients with adjuvant trastuzumab therapy. *Springerplus*. 2014;3:620.
28. <https://www.fda.gov/files/drugs/published/Qualification-Process-for-Drug-Development-Tools.pdf>
29. Kosiborod MN, Cherney DZI, Desai AS, et al. Sodium zirconium cyclosilicate for management of hyperkalemia during spironolactone optimization in patients with heart failure. *J Am Coll Cardiol*. 2025;85:971–984.
30. Borlaug BA, Testani JM, Petrie MC, et al. Effects of vardenafil in worsening heart failure with preserved ejection fraction: a phase 2 randomized trial. *Nat Med*. 2025;31(11):3853–3861.
31. Fontana M, Solomon SD, Kachadourian J, et al. CRISPR-Cas9 gene editing with nexiguran zicmuran for ATTR cardiomyopathy. *N Engl J Med*. 2024;391:2231–2241.
32. Gillmore JD, Gane E, Taubel J, et al. CRISPR-Cas9 in vivo gene editing for transthyretin amyloidosis. *N Engl J Med*. 2021;385:493–502.
33. Fleming TR, DeMets DL. Surrogate end points in clinical trials: are we being misled? *Ann Intern Med*. 1996;125:605–613.
34. Revkin J. Framework for defining evidentiary standards for surrogate endpoints. Accessed December 1, 2025. <https://fnih.org/wp-content/uploads/2023/06/5-Revkin-LDL-HDL-public-1.pdf>
35. Savarese G, Uijl A, Ouwerkerk W, et al. Biomarker changes as surrogate endpoints in early-phase trials in heart failure with reduced ejection fraction. *ESC Heart Fail*. 2022;9:2107–2118.
36. U.S. Food and Drug Administration (FDA). *Surrogate Endpoint Resources for Drug and Biologic Development*. FDA; 2018.
37. Table of surrogate endpoints that were the basis of drug approval or licensure. 2025. *Food and Drug Administration*. 2025. Accessed December 1, 2025. <https://www.fda.gov/drugs/development-resources/table-surrogate-endpoints-were-basis-drug-approval-or-licensure>
38. Enrichment strategies for clinical trials to support approval of human drugs and biological products. 2019. Food and Drug Administration. 2019. Accessed December 1, 2025. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/enrichment-strategies-clinical-trials-support-approval-human-drugs-and-biological-products>
39. Schmitt W, Rühls H, Burghaus R, et al. NT-proBNP qualifies as a surrogate for clinical endpoints in heart failure. *Clin Pharmacol Ther*. 2021;110:498–507.
40. Odutayo A, Cherney DZI. Predictive enrichment in kidney RCTs: is albuminuria the answer? *J Am Soc Nephrol*. 2021;32:2689–2691.
41. van der Leeuw J, Oemrawsingh RM, van der Graaf Y, et al. Prediction of absolute risk reduction of cardiovascular events with perindopril for individual patients with stable coronary artery disease - results from EUROPA. *Int J Cardiol*. 2015;182:194–199.
42. Karagiannis T, Tsapas A, Bekieri E, Toulis KA, Nauck MA. A methodological framework for meta-analysis and clinical interpretation of subgroup data: the case of major adverse cardiovascular events with GLP-1 receptor agonists and SGLT2 inhibitors in type 2 diabetes. *Diabetes Care*. 2024;47:184–192.
43. Beydoun MA, Noren Hooten N, Georgescu MF, et al. Serum neurofilament light chain as a prognostic marker of all-cause mortality in a national sample of US adults. *Eur J Epidemiol*. 2024;39:795–809.
44. White HD, Held C, Stewart R, et al. Darapladib for preventing ischemic events in stable coronary heart disease. *N Engl J Med*. 2014;370:1702–1711.
45. O'Donoghue ML, Braunwald E, White HD, et al. Effect of darapladib on major coronary events after an acute coronary syndrome: the SOLID-TIMI 52 randomized clinical trial. *JAMA*. 2014;312:1006–1015.
46. Gupta DK, de Lemos JA, Ayers CR, Berry JD, Wang TJ. Racial differences in natriuretic peptide levels: the Dallas Heart Study. *JACC Heart Failure*. 2015;3:513–519.
47. Ibrahim NE, Burnett JC Jr, Butler J, et al. Natriuretic peptides as inclusion criteria in clinical

trials: a JACC: heart failure position paper. *JACC Heart Failure*. 2020;8:347-358.

48. O'Hanlon R, O'Shea P, Ledwidge M, et al. The biologic variability of B-type natriuretic peptide and N-terminal pro-B-type natriuretic peptide in stable heart failure patients. *J Card Fail*. 2007;13:50-55.

49. Aylward LL, Hays SM, Smolders R, et al. Sources of variability in biomarker concentrations. *J Toxicol Environ Health B Crit Rev*. 2014;17:45-61.

50. Zaninotto M, Padoan A, Mion MM, Marinova M, Plebani M. Short-term biological variation and diurnal rhythm of cardiac troponin I (Access hs-TnI) in healthy subjects. *Clin Chim Acta*. 2020;504:163-167.

51. Schimmel H, Zegers I, Emons H. Standardization of protein biomarker measurements: is it feasible? *Scand J Clin Lab Invest Suppl*. 2010;242:27-33.

52. Apple FS, Jaffe AS, Collinson P, et al. IFCC educational materials on selected analytical and clinical applications of high sensitivity cardiac troponin assays. *Clin Biochem*. 2015;48:201-203.

53. Aimo A, Sanguinetti C, Franzini M. Screening for cardiac involvement in carriers of pathogenic TTR variants: proposal for an approach based on high-sensitivity troponin. *Circulation*. 2024;150:1656-1658.

54. Analytical procedures and methods validation for drugs and biologics. 2015. *Food and Drug Administration*. 2015. Accessed December 1, 2025. <https://www.fda.gov/files/drugs/published/Analytical-Procedures-and-Methods-Validation-for-Drugs-and-Biologics.pdf>

55. Heerspink HJL, Stefánsson BV, Correa-Rotter R, et al. Dapagliflozin in patients with chronic kidney disease. *N Engl J Med*. 2020;383:1436-1446.

56. Brenner BM, Cooper ME, de Zeeuw D, et al. Effects of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. *N Engl J Med*. 2001;345:861-869.

57. Cunningham JW, Vaduganathan M, Claggett BL, et al. Effects of sacubitril/valsartan on N-terminal pro-B-type natriuretic peptide in heart failure with preserved ejection fraction. *JACC Heart Failure*. 2020;8:372-381.

58. Lobo R, De Michieli L, Jaffe AS. Sex-specific 99th percentile URLs for cardiac troponin assays-their time has come. *Clin Chem*. 2021;67:197-200.

59. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371:993-1004.

60. Solomon SD, McMurray JJV, Anand IS, et al. Angiotensin-neprilysin inhibition in heart failure with preserved ejection fraction. *N Engl J Med*. 2019;381:1609-1620.

61. Armstrong PW, Pieske B, Anstrom KJ, et al. Vericiguat in patients with heart failure and reduced ejection fraction. *N Engl J Med*. 2020;382:1883-1893.

62. Kondo T, Campbell R, Jhund PS, et al. Low natriuretic peptide levels and outcomes in patients with heart failure and preserved ejection fraction. *JACC Heart Failure*. 2024;12:1442-1455.

63. Butt JH, Adamson C, Docherty KF, et al. Efficacy and safety of dapagliflozin in heart failure with reduced ejection fraction according to N-terminal pro-B-type natriuretic peptide: insights from the DAPA-HF trial. *Circ Heart Fail*. 2021;14:e008837.

64. Ezekowitz JA, O'Connor CM, Troughton RW, et al. N-terminal pro-B-type natriuretic peptide and clinical outcomes: Vericiguat Heart Failure With Reduced Ejection Fraction Study. *JACC Heart Failure*. 2020;8:931-939.

65. Hicks KA. Endpoints in cardiovascular trials. 2022. *Food and Drug Administration*. 2022. Accessed December 1, 2025. <https://www.fda.gov/media/167530/download>

66. *Clinical investigation of medicinal products for the treatment of acute heart failure - scientific guideline*. European Medicines Agency; 2015. Accessed December 1, 2025. <https://www.ema.europa.eu/en/clinical-investigation-medicinal-products-treatment-acute-heart-failure-scientific-guideline>

67. Esquivel Gaytan A, Bomer N, Grote Beverborg N, van der Meer P. 404-error "Disease not found": unleashing the translational potential of -omics approaches beyond traditional disease classification in heart failure research. *Eur J Heart Fail*. 2024;26:1313-1323.

68. Bayes-Genis A, Liu PP, Lanfear DE, et al. Omics phenotyping in heart failure: the next frontier. *Eur Heart J*. 2020;41:3477-3484.

69. Cleland JGF, Ferreira JP, Mariotti B, et al. The effect of spironolactone on cardiovascular function and markers of fibrosis in people at increased risk of developing heart failure: the Heart 'Omics' in AGEing (HOMAGE) randomized clinical trial. *Eur Heart J*. 2021;42:684-696.

70. Ouwerkerk W, Belo Pereira JP, Maasland T, et al. Multiomics analysis provides novel pathways related to progression of heart failure. *J Am Coll Cardiol*. 2023;82:1921-1931.

71. Shah ASV, Keene SJ, Pennells L, et al. Cardiac troponins and cardiovascular disease risk prediction: an individual-participant-data meta-analysis. *J Am Coll Cardiol*. 2025;85:1471-1484.

72. Ridker PM, Danielson E, Fonseca FA, et al. Reduction in C-reactive protein and LDL cholesterol and cardiovascular event rates after initiation of rosuvastatin. *Lancet*. 2009;373:1175-1182.

73. Neumann JT, de Lemos JA, Apple FS, Leong DP. Cardiovascular biomarkers for risk stratification in primary prevention. *Eur Heart J*. 2025;46:3823-3843.

74. Jia X, Sun W, Hoogeveen RC, et al. High-sensitivity troponin I and incident coronary events, stroke, heart failure hospitalization, and mortality in the ARIC study. *Circulation*. 2019;139:2642-2653.

75. Jia X, Al Rifai M, Ndumele CE, et al. Reclassification of pre-heart failure stages using cardiac biomarkers: the ARIC study. *JACC Heart Failure*. 2023;11:440-450.

76. Sandoval Y, Bielinski SJ, Daniels LB, et al. Atherosclerotic cardiovascular disease risk stratification based on measurements of troponin and

coronary artery calcium. *J Am Coll Cardiol*. 2020;76:357-370.

77. Rahsepar AA, Bluemke DA, Habibi M, et al. Association of pro-B-type natriuretic peptide with cardiac magnetic resonance-measured global and regional cardiac function and structure over 10 years: the MESA study. *J Am Heart Assoc*. 2021;10:e019243.

78. Apple FS, Sandoval Y, Jaffe AS, Ordóñez-Llanos J. Cardiac troponin assays: guide to understanding analytical characteristics and their impact on clinical care. *Clin Chem*. 2017;63:73-81.

79. Ridker PM, Everett BM, Thuren T, et al. Antiinflammatory therapy with canakinumab for atherosclerotic disease. *N Engl J Med*. 2017;377:1119-1131.

80. Everett BM, Brooks MM, Vlachos HE, Chaitman BR, Frye RL, Bhatt DL. Troponin and cardiac events in stable ischemic heart disease and diabetes. *N Engl J Med*. 2015;373:610-620.

81. Stone GW, Sabik JF, Serruys PW, et al. Everolimus-eluting stents or bypass surgery for left main coronary artery disease. *N Engl J Med*. 2016;375:2223-2235.

82. Mäkilä T, Holm NR, Lindsay M, et al. Percutaneous coronary angioplasty versus coronary artery bypass grafting in treatment of unprotected left main stenosis (NOBLE): a prospective, randomised, open-label, non-inferiority trial. *Lancet*. 2016;388:2743-2752.

83. Gaudio M, Jaffe AS, Milojevic M, et al. Great debate: myocardial infarction after cardiac surgery must be redefined. *Eur Heart J*. 2024;45:4170-4177.

84. Palladini G, Campana C, Klersy C, et al. Serum N-terminal pro-brain natriuretic peptide is a sensitive marker of myocardial dysfunction in AL amyloidosis. *Circulation*. 2003;107:2440-2445.

85. Maurer MS, Dunnmon P, Fontana M, et al. Proposed cardiac end points for clinical trials in immunoglobulin light chain amyloidosis. *Circ Heart Fail*. 2022;15:e009038.

86. Blanco-Dominguez R, Sánchez-Díaz R, de la Fuente H, et al. A novel circulating microRNA for the detection of acute myocarditis. *N Engl J Med*. 2021;384:2014-2027.

87. Staerk L, Preis SR, Lin H, et al. Protein biomarkers and risk of atrial fibrillation: the FHS. *Circ Arrhythm Electrophysiol*. 2020;13:e007607.

88. Hussein AA, Bartz TM, Gottdiener JS, et al. Serial measures of cardiac troponin T levels by a highly sensitive assay and incident atrial fibrillation in a prospective cohort of ambulatory older adults. *Heart Rhythm*. 2015;12:879-885.

89. van den Berg NWE, Neefs J, Kawasaki M, et al. Extracellular matrix remodeling precedes atrial fibrillation: results of the PREDICT-AF trial. *Heart Rhythm*. 2021;18:2115-2125.

90. Hijazi Z, Siegbahn A, Andersson U, et al. High-sensitivity troponin I for risk assessment in patients with atrial fibrillation: insights from the Apixaban for Reduction in Stroke and other Thromboembolic Events in Atrial Fibrillation (ARISTOTLE) trial. *Circulation*. 2014;129:625-634.

91. Hijazi Z, Wallentin L, Siegbahn A, et al. High-sensitivity troponin T and risk stratification in patients with atrial fibrillation during treatment

with apixaban or warfarin. *J Am Coll Cardiol*. 2014;63:52–61.

92. U.S. Food and Drug Administration (FDA). *Expedited Programs for Serious Conditions – Drugs and Biologics. Guidance for Industry*. FDA; 2014.

93. European Medicines Agency. Qualification of novel methodologies for drug development: guidance to applicants. EMA/CHMP/SAWP/72894/

2008 Rev. 5. Accessed December 1, 2025. https://www.ema.europa.eu/en/documents/regulatory-procedural-guideline/qualification-novel-methodologies-drug-development-guidance-applicants_en.pdf

94. Elliott P, Cowie MR, Franke J, et al. Development, validation, and implementation of biomarker testing in cardiovascular medicine state-of-the-art. *Cardiovasc Res*. 2021;117:1248–1256.

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APPENDIX For supplemental tables, please see the online version of this paper.